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A hybrid multi-objective evolutionary algorithm for solving an adaptive flexible job-shop rescheduling problem with real-time order acceptance and condition-based preventive maintenance

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ABSTRACT

Production scheduling and maintenance planning are two of the most important tasks in the modern manufacturing workshop. Meanwhile, due to the dynamic order arrival and real-time machine monitoring information updating, the integrated optimization of them becoming more complex and meaningful. Therefore, this study intends to address an adaptive flexible job-shop rescheduling problem with real-time order acceptance (ROA) and condition-based preventive maintenance (CBPM). More precisely, the main innovative works are described as follows: (1) a CBPM policy with both imperfect preventive maintenance (PM) and four inspection strategies is designed to find the optimal maintenance planning for each production machine; (2) a multi-objective optimization model is developed for the concerned problem; and (3) a hybrid multi-objective evolutionary algorithm (HMOEA) with hybrid initialization method, hybrid local search operators and adaptive rescheduling strategies is proposed. In the numerical simulation, the performance and competitiveness of the proposed CBPM policy are first demonstrated by comparing with other maintenance policies. Second, the effectiveness and superiority of parameter setting, order sorting rules, improved operators and overall performance of the proposed algorithm are verified by internal analysis of the algorithm. Third, an adaptive rescheduling strategy pool is constructed by running three rescheduling strategies on all rescheduling scenarios. Finally, a comprehensive sensitivity analysis is performed to illustrate the impact of several critical parameters on the adaptive rescheduling problem, and the results and comparisons show that the proposed HMOEA algorithm and order acceptance strategy have good robustness in most parameters.

1. Introduction

With the individualization of market demand, the production mode of manufacturing firms is gradually changing from make-to-stock (MTS) to make-to-order (MTO) or customer-oriented production (Chen et al., 2021) since some products cannot be manufactured until identifying the specific customer needs (Federgrue & Katalan, 1999). Under this context, quickly response to customer demands have received a great deal of attention from academia and industry due to its indispensable roles in enhancing customer satisfaction and decreasing production lead time. Furthermore, according to the production experience, accepting and processing all potential orders may not be a shrewd decision for a MTO firm due to the limited production capacity and strict delivery due date, which may result in reduced revenue and even lost customers (Li et al., 2018). Therefore, order acceptance and scheduling (OAS) has attracted considerable attention (Chaurasia & Singh, 2017; Yin et al., 2016) and should go along with analyzing the real-time capacity utilization.

Nevertheless, most of OAS studies assume that the machines are continuously available during the planning horizon, which is a serious deviation from the actual production since a mass of conditions (e.g., breakdown) may cause the machine unavailable. It is worth noting that, maintenance has been recognized as an effective method for keeping machine performance and improving production efficiency (Ghaleb et al., 2020), and an appropriate maintenance planning can save 20%–30% of the total operation cost (Syan & Ramsoobag, 2019). Thus, many scholars and practitioners focus on the

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reliability and maintainability, and a lot of maintenance strategies are proposed (de Jonge & Scarf, 2020), such as time-based maintenance (TBM) and condition-based maintenance (CBM). Among them, TBM ignores the actual production conditions, resulting in the formulated maintenance planning is hard or even impossible to execute in practice. However, with the repaid development of advanced sensor and information processing technologies, CBM is more suitable for modern production since it can recommend maintenance actions based on the information collected through condition monitoring (de Jonge et al., 2017; Jardine et al., 2006). Meanwhile, modeling continuous degradation as a stochastic process has been extensively studied in the literature, such as Markov chain, Gamma process, Inverse Gaussian (IG) process and Wiener process (Ye & Chen, 2014). Among them, the Gamma process was firstly proposed by Abdel-Hameed (1975), which has drawn a lot of attentions due to the nice mathematical properties and meaningful physical interpretations (Alaswad & Xiang, 2017; Ye & Xie, 2015).

As far as we know, production scheduling and maintenance planning, as two key elements, have gained more and more concerns from the community of operational research since they aim to increase not only the productivity but also some indicators (Zandieh et al., 2017), and many real-world scheduling problems usually involve these two elements, such as in job shop (Wang & Yu, 2010) and high-speed railway (Zhang et al., 2019). According to the previous research findings (An et al., 2021; Xiao et al., 2020), the hierarchical optimization effect of production scheduling and maintenance planning is obviously inferior to the integrated optimization, so the integrated optimization has become one of the research hot topics. In order to utilize the real-time monitoring information, some scholars have begun to simulate the operation state by using numerical method, such as the work of Zandieh et al. (2017), Rahmati et al. (2018), Ghaleb et al. (2020) and Sharifi and Taghipour (2021). However, the simulation technology has some drawbacks, e.g., uncertainty and randomness. Therefore, we formulate the maintenance model by analytically expression, and a discrete approximate iteration (DAI) algorithm is designed to optimize the maintenance planning for each production machine.

Furthermore, previous research is mainly concentrated in a static environment, rather than in a dynamic environment. To fill this gap, we are dedicated to deal with an adaptive flexible job-shop rescheduling problem with real-time order acceptance (ROA) and condition-based preventive maintenance (CBPM). More specifically, the main contributions of this study are fivefold: (1) a CBPM policy with both imperfect preventive maintenance (PM) and four inspection policies is designed to find the optimal maintenance planning for each machine, and the superiority is demonstrated by comparing with other traditional maintenance policies; (2) a multi-objective optimization model is developed for the adaptive rescheduling problem to minimize the makespan, instability and cumulative production cost; (3) a hybrid multi-objective evolutionary algorithm (HMOEA) with several improved operators is proposed to deal with the problem, and the effectiveness and superiority are demonstrated by comparing with other well-known intelligent algorithms; (4) an adaptive rescheduling strategy pool is constructed by running all candidate rescheduling strategies on all rescheduling scenarios; and (5) a comprehensive sensitivity analysis is provided to illustrate the impact of several key parameters (including the initial order acceptance ratios, rejection penalty and instability threshold) on the adaptive rescheduling problem.

The rest of this paper is organized as follows. Section 2 provides a comprehensive literature review and detailed research gaps. Section 3 describes the concerned problem, and formulates the CBPM policy and a multi-objective optimization model. Section 4 presents the proposed HMOEA algorithm. In Section 5, the performance of the proposed CBPM policy is verified. Next, the effectiveness and superiority of the designed internal operators and overall performance of the presented algorithm are demonstrated in Section 6. After that, an adaptive rescheduling strategy pool is constructed in Section 7. Section 8 elaborates the effect of several key parameters on the adaptive rescheduling problem. Finally, some conclusions and perspectives for future work are given in Section 9.

2. Literature review and research gaps

This work is related to three research streams: (1) order acceptance and (re)scheduling; (2) integrated optimization of production scheduling and maintenance planning; and (3) multi-objective evolutionary algorithms (MOEAs). To be more intuitive, the related publications will be reviewed based on the three items.

2.1. Order acceptance and (re)scheduling

The OAS problem was firstly investigated by Bartal et al. (1996), who investigated the problem of minimizing the weighted sum of the makespan and rejection penalties on identical parallel machines. Since then, a large number of scholars and practitioners have focused on this subject. For example, Lewis and Slotnick (2002) extended the classic OAS problem to multiple planning periods for the case that rejecting an order of a customer will result in the loss of all the future orders from that customer. Slotnick and Morton (2007) focused on order acceptance when the production capacity is limited, and the customers can receive a discount for the late delivery. Lin and Ying (2015) aimed to maximize the total net profit for a permutation flow shop scheduling problem with order acceptance and weighted tardiness. Xiao et al. (2015) studied OAS problem in non-permutation flow shop to minimize the makespan. Naderi and Roshanaei (2020) dealt with the order acceptance and identical parallel machine scheduling problem by employing a Bender-based decomposition approach. Wang, Geng, and Cheng (2018), Wang et al. (2015) investigated the OAS problem in subcontracting context, which aimed to maximize the profit. Xiao et al. (2010) and Xu and Chen (2019) researched the OAS problem with different customer priorities, e.g., high-level and low-level customers. For a more detailed discussion about the OAS and its variants, the reader can refer to Shabtay et al. (2013), Volling et al. (2013) and Leng et al. (2021) and the references therein.

However, the foregoing papers mainly focused on the deterministic situation, neglecting the effect of dynamic order arrives. To fill this gap, several studies considered future order arrivals in order acceptance decision and defined the problem as a dynamic OAS problem. For instance, Ebben et al. (2005) proposed several workload-based order acceptance strategies with stochastic order arrivals. Arredondo and Martinez (2010) developed a novel approach for a dynamic order acceptance policy, in which the average-reward reinforcement learning was designed. van Foreest et al. (2010) assumed that the arriving orders follow an independent stationary Poisson process and several OAS policies were compared by simulation. Xu et al. (2015) formulated a dynamic programming model by considering the stochastic arrival orders and sequence-dependent setup times (SDST). Rahman et al. (2015) focused on the ROA and rescheduling strategies for permutation flow shop scheduling problems (PFSPs) to separately optimize the makespan and profit. Wang et al. (2017) considered a multi-objective dynamic scheduling problem in response to continuous arrivals of new jobs under the assumption that the job processing time can be controlled by allocating extra resource. Wang, Yin, and Cheng (2018) concentrated on the rescheduling problem on identical parallel machines in the presence of unexpected delays that lead some jobs unavailable, and the main objective was to minimize the makespan, while keeping other two objective values no greater than their limits. Jiang et al. (2020) developed a dynamic OAS model that aimed to maximize the total profit. Chen et al. (2021) studied a joint optimization problem for ROA and job release planning in a permutation flow shop with data-driven production status and job makespan prediction. An, Chen, Gao, Zhang, et al. (2022) addressed an integrated optimization problem for ROA and flexible job-shop rescheduling with multi-level imperfect PM constraints, in which the order can be rejected with a certain penalty.

Although the aforementioned studies have done a lot of research on OAS problem, most of them focus on static OAS problem and simple environments (e.g., single machine, parallel machine and flow shop),

while few work has paid attention to both the on-line order acceptance and dynamic flexible job-shop rescheduling. Furthermore, in practice, the optimal schedule can easily be interrupted by many unexpected events (e.g., new job arrivals and machine breakdowns), which render the existing schedule no longer optimal or even infeasible (Wang, Yin, & Cheng, 2018). Therefore, rescheduling is inevitable, and the predictive-reactive scheduling is widely applied in practice compared to other scheduling strategies, e.g., completely reactive scheduling and robust scheduling (Ghaleb et al., 2021). Meanwhile, to the best of our knowledge, the objective of existing studies mainly concerns maximizing the number of accepted orders and total profit, or minimizing the order completion time and production cost (Jiang et al., 2017; Rahman et al., 2015). Nevertheless, they almost ignored the impact of rescheduling on the original or previous scheduling plan, i.e., instability. Under these contexts, we extend this stream literature by the following two ways: (1) adaptive rescheduling methods (i.e., according to the task sizes of the inserted order and the remaining task size to select an appropriate response strategy) is designed to deal with the ROA and flexible job-shop rescheduling problem with instability and availability constraints; and (2) an order addition heuristic method (Liu, 2019) is employed to generate a smaller initial order acceptance jobset to alleviate the computational burden in each rescheduling point.

2.2. Integrated optimization of production scheduling and maintenance planning

In real-life situation, deterioration and failure of production systems or devices are commonplace and inevitable with the increased usage and age. To improve the machine performance and increase the productivity, integrated optimization of the production scheduling and maintenance planning is one of the research hotspots in recent years. Generally speaking, the integrated models can be broadly classified into two types (Ghaleb et al., 2021), i.e., integrated models with TBM activities and integrated models with CBM activities.

Regarding integrated models with TBM activities, a large number of studies have been emerged. For example, Wang and Yu (2010), Li et al. (2014) and An et al. (2021) dealt with multi-objective flexible job-shop scheduling problem (MOFJSP) with maintenance time window constraints. Baykasoğlu and Madenoğlu (2021) proposed a greedy randomized adaptive search procedure (GRASP) for the integrated scheduling with fixed or flexible PM. Huo and Zhao (2018) studied two machine scheduling problem subject to arbitrary machine availability constraint. Cui and Lu (2017) coped with a single machine scheduling problem with flexible periodic PM and the maximum continuous working time was restricted. However, in reality, the maintenance planning should be determined by the actual working conditions, rather than given in advance (i.e., predefined maintenance). Thus, the studies considering the actual operation time or conditions have gradually emerged to remedy this defect. For instance, Han et al. (2019) addressed the flow shop rescheduling problem with machine breakdowns, in which the breakdown time and maintenance duration were subjected to a uniform distribution. Liu, Zhang, et al. (2021) focused on the integrated optimization of production planning and PM scheduling on synchronized parallel machines, and the uncertainties of machine failure, completion time and minimal repair (MR) were considered. Chen, An, et al. (2020) devised an approximate algorithm to jointly optimize production scheduling and maintenance planning based on the reliability intervals which were divided by two reliability thresholds. An et al. (2020) simultaneously optimized production scheduling and imperfect cutting tool maintenance considering the total energy consumption. Recently, Xiao et al. (2020) focused on flow shop production scheduling and maintenance planning with varying operational conditions. Boufellouh and Belkaid (2020) explored the integrated production scheduling and maintenance planning in a permutation flow shop, and the scarcity of a global resource was taken into account to minimize the makespan and total production cost. More recently, An, Chen, Gao, Li, and Zhang (2022) employed three rescheduling strategies to cope with the joint optimization problem with new job insertion and machine PM. An, Chen, Gao, Zhang, et al. (2022) dealt with an integrated optimization problem of flexible job-shop rescheduling and multi-level imperfect maintenances, i.e., MR, PM, overhaul maintenance (OM) and replacement. An, Chen, Hu, et al. (2022) studied an integrated optimization problem of PM and flexible job-shop rescheduling with new machine insertion and processing speed selection.

With the rapid development of prognostics and health management (PHM) technologies, CBM activities and machine degradation descriptions are gradually becoming feasible, and the researchers usually employ a simulation-based optimization method to model machine's deterioration and arrange maintenance actions. For example, Grall et al. (2002) proposed a perfect maintenance policy to optimize the preventive replacement (PR) threshold and the inspection schedule based on the system state in order to balance the cost engaged by failure and unavailability on an infinite horizon. Zandieh et al. (2017) considered a simulation model with a combination of Sigmoid function and Gaussian distribution to simulate the CBM, and then, the maintenance scheme was entered into the flexible scheduling plan to optimize the makespan by improved imperialist competitive algorithm. Rahmati et al. (2018) constructed an integrated optimization model for CBM and FJSP to address the corrective replacement (CR) and PR problem, in which they assumed that the machine degradation was only affected by the discrete shock. Ghaleb et al. (2020) and Sharifi and Taghipour (2021) developed an integrated mathematical model for production scheduling and maintenance planning in a degrading single machine environment, and the expected values of maintenance parameters were derived by Markov process or chain. Pan et al. (2012) developed a single-machine-based integrated scheduling model with the aim of minimizing the tardiness by balancing the production scheduling and predictive maintenance planning. Yang et al. (2021) utilized reinforcement learning (RL) algorithm to integrate optimization of PM and production scheduling in a multi-state single-machine production system with deterioration effects. Mokhtari and Dadgar (2015) developed a mixed integer linear programming (MILP) model to represent the FJSP under PM, and Monte Carlo Simulation (MCS) was used to minimize the total number of tardy jobs after PM. It is worth noting that, Ghaleb et al. (2021) aimed at solving the joint optimization problem of production scheduling and CBM planning in a flexible job shop with real-time dynamics, e.g., new job arrivals and machine breakdowns. In their research, a matrix-based approach was used to estimate the deterioration, and the processing time and production costs were affected by the condition.

The above-mentioned research mainly focused on the static scheduling and PM, and there was few research on both the real-time OAS and CBPM optimization. To fill this gap, some investigators have begun to study it, but most of the maintenance models are based on the failure-time data. For instance, Iranpoor et al. (2014) investigated a novel extensive problem of order acceptance, scheduling and due-date assignment together with machine modifying maintenance decision. Zhong et al. (2014) established an OAS model with machine availability constraints, specifically, the machine can be only available within a number of discontinuous time intervals. Rahman et al. (2018) proposed a rescheduling approach to deal with both single and multiple orders while considering random machine interruptions. In order to further enrich the integrated optimization of production scheduling and maintenance planning, we contribute to this literature stream by the following four ways: (1) a dynamic integrated optimization problem is studied, more precisely, the number of CBMs, PM threshold and maximum inspection interval are unknown and should be jointly determined as decision variables with order acceptance and scheduling; (2) an analytical derivation is performed to characterize the expected cost rate function in CBM instead of solving it directly through numerical simulation; (3) a periodic inspection and three non-periodic inspection strategies are designed to detect the degraded state of the production machine, since discrete inspections can be more appropriate

and cost effective for some practical engineering systems (Alebrant Mendes et al., 2017; Hao et al., 2020; Nguyen et al., 2019); and (4) the expected total cost of each machine in a renew cycle is considered to be minimized, that consists of the inspection cost, maintenance cost, downtime cost and operating cost.

2.3. Multi-objective evolutionary algorithms

According to the forgoing description, the concerned problem in this paper is also an NP-hard problem, since it already includes classical OAS and FJSP, which are both NP-hard problems (Blazewicz et al., 1988; Ghosh, 1997). To cope with this kind of problem, heuristic and meta-heuristic algorithms have become the preferred techniques (Chaurasia & Singh, 2017; Gao, Cao, et al., 2019; Gao et al., 2014; Wang et al., 2013) since they can obtain an optimal or a sub-optimal solution within a reasonable computational time. Until now, a variety of MOEAs have been proposed, e.g., NSGA-II (Deb et al., 2002), MOEA/D (Zhang & Li, 2007) and NSGA-III (Deb & Jain, 2014). Among them, NSGA-III was presented by Deb and Jain (2014), and the superiority has been demonstrated by Deb and Jain (2014) and Yuan et al. (2015). Since then, based on the standard NSGA-III, a large number of variants have been designed. For example, Cheng et al. (2016) proposed a reference vector-guided evolutionary algorithm (RVEA), and an angle-penalized distance (APD) was designed to balance the convergence and diversity of the solutions in the high-dimensional objective space. Xiang et al. (2017) employed a maximum-vector-angle-first principle to guarantee the wideness and uniformity of the solutions. Liu, Jin, et al. (2021) designed an adaptive RVEA by using an improved growing neural gas to deal with many-objective irregular optimization problems. An, Chen, Hu, et al. (2022) proposed a bi-population cooperative evolutionary algorithm with improved APD (i.e., IAPD) individual elimination strategy and enhanced genetic operators. According to the research of Deb et al. (2002), He and Yen (2017) and An, Chen, Gao, Li, and Zhang (2022), the Pareto rule is still an effective method in solving low-dimensional many-objective optimization problems (MaOPs). Thus, we also use the Pareto rule for individual sorting in this paper. Meanwhile, through analyzing the MOEAs, it is found that many algorithms are designed on the basis of genetic algorithm (GA). Since the local search (LS) capability of standard GA is insufficient (Li et al., 2019), there are amounts of LS methods are presented (Gao et al., 2015; Li & Gao, 2016) to enhance the performance, such as variable neighborhood search (VNS) and tabu search (TS). According to the research findings of Nowicki and Smutnicki (1996) and Balas and Vazacopoulos (1998), exchanging two adjacent internal critical operations in critical operation block and permuting non-critical operations cannot decrease the makespan. Therefore, VNS based on critical path (e.g., N5 structure) is widely adopted by many scholars and the superiority has been verified (Balas & Vazacopoulos, 1998). Besides, the orthogonal learning (OL) strategy is another effective method, and the research by Chen, Heidari, et al. (2020) and An, Chen, Hu, et al. (2022) show that OL has significant advantages in solving the integrated optimization problem. In addition, not all Pareto fronts (PFs) are regular in practical engineering problems, irregular PFs are a more common phenomenon (Xiang et al., 2020). As a consequence, a periodic adjustment strategy is designed by Cheng et al. (2016) to update the reference vector (RV) according to the distribution of solutions. However, in previous research (An, Chen, Gao, Li, & Zhang, 2022; An, Chen, Gao, Zhang, et al., 2022), we found that non-periodically changing RV may have good adaptability by comparing with periodically changing RV, which is also applied in this study.

In general, although a lot of research have been devoted to improving the performance of MOEAs, our research has the following differences from previous studies: (1) a hybrid local search with N5 structure, processing time window and OL strategy is constructed, more specifically, N5 structure is applied to swap the first two and last two operations on the critical block (i.e., change the processing sequence on the original machine), the processing time window is utilized to move the internal operations in the critical block to other selectable machines (i.e., change the processing machine), and the OL strategy is applied in production scheduling to speed up the search quality for the optimal machine combinations of the selected operations, i.e., the optional machines of the selected operations constitute the orthogonal table; (2) a hybrid initialization method (named as GCC) is designed according to the characteristics of the concerned problem to improve the quality of the initial population; (3) the TOPSIS method (Tavana et al., 2016) is employed to choose the individual with the highest score in the iteration for genetic manipulation (Zhang et al., 2020); (4) an adaptive response program including three rescheduling strategies is designed to realize parameter recording and transmission under different rescheduling strategies, as well as overall performance evaluation; and (5) an optimal rescheduling response strategy pool is developed under different rescheduling scenarios to find the best production-maintenance planning. For easy expression, the hybrid LS, hybrid initialization method, individual selection method and adaptive rescheduling strategies are all incorporated into the standard NSGA-III, labeled as HMOEA.

3. Problem description and mathematical model

In this section, the assumptions and notations are first introduced. Second, the concerned problem is formally defined. Third, a CBPM policy is proposed to formulate the optimal maintenance planning for each independent machine. Next, a multi-objective optimization model is developed. Finally, a simple illustrative example is presented to explain the problem.

3.1. Assumptions and notations

The concerned problem is based on the following assumptions:

(1) The firm has the right to accept or reject any incoming order. Once an order or a job is accepted, it cannot be rejected later.

(2) The release date and processing route of each job, and the processing time of each operation on the candidate machine set are known when a new job is inserted, and the remaining operations of the original scheduling plan may be rescheduled when a new job arrives.

(3) Each operation can only be processed by one candidate machine at one time, and if one operation is interrupted, its remaining part will be processed first after the machine is restored.

(4) All machines are ready at time zero, and each machine can only process one operation at a time and the degradation can be characterized by Gamma process.

(5) The degradation state and level of the machine can only be detected by discrete inspection, and the inspection is instantaneous, perfect and non-destructive, but incurs a cost $C_I > 0$.

(6) A CR action will cost more than a PR action, but CR and PR consume the same time. Similarly, a perfect PM action will cost much more than an inspection, i.e., $C_{\mathrm{CR}} > C_{\mathrm{PR}} > C^0_{\mathrm{PM}} > C_I$. Besides, the maintenance resources (e.g., spare parts and repairmen) are sufficient.

(7) The degradation level after maintenance (Z_l) and the degradation level before maintenance (X_l) are independent, and Z_l is also a random variable and no greater than the PM threshold ξ (i.e., $Z_l \leq \xi$).

Also, the used notations in this study are summarized in Table 1.

3.2. Problem description

The concerned problem can be formally defined as follows: suppose that there is a set of N_A jobs $J = \{J_1, J_2, \ldots, J_{N_A}\}$ is already accepted, which exist in a flexible job shop before new orders arrive. The job shop contains K different machines $M = \{M_1, M_2, \ldots, M_K\}$, and each machine may have different processing time and cost for the same operation. At rescheduling time t', a set of orders come

Table 1
Notations.

Symbol	Definition
Indices	
i, i'	Index for jobs, $\{i, i'\} \subseteq \{1, 2, \ldots, N_A + N_N + N_R\}$
j, j'	Index for operations, $\{j, j'\} \subseteq \{1, 2, \ldots n_i\}$
k, k'	Index for machines, $\{k, k'\} \subseteq \{1, 2, \ldots, K\}$
l, m	Index for maintenance cycles
r, s	Index for inspections in each maintenance cycle
o/c	Index for inspection policies/rescheduling times
Parameters	
N_A/N_N	Total number of orders already accepted/newly accepted orders
K/N_R	Total number of machines/rejected orders
N_0	Total number of operations in the last rescheduling moment
N_0'	Total number of remaining unstarted operations in previous scheduling plan
N_A'	Total number of jobs with remaining unstarted operations in current time t'
N_k	Total number of maintenance times for machine k in the scheduling horizon
$Cycle/Q$	Total number of scheduling times/operations in each test instance
O_{ij}	The jth operation of job i
R_i	The release time of order i
n_i/n_i'	Total number of operations/remaining unstarted operations of order i
$O_{i'j'}$	Immediate predecessor activity of O_{ij} on the same machine
S_{ijk}/P_{ijk}	The start/normal processing time of operation O_{ij} on machine M_k
$E_{ijk}/E_{i'j'k}$	End processing time of operation $O_{ij}/O_{i'j'}$ on machine M_k
AMS_{ij}	The alternative or available machine set of operation O_{ij}
$E^c(O_{ij})/L^s(O_{ij})$	The earliest completion time/latest start time of operation O_{ij}
C_i^{com}	Completion time of order i in current scheduling scheme
γ_0/γ_c	Initial ratio of the accepted orders at time zero/at the cth scheduling point ($c > 1$)
λ	Rejection loss factor for each rejected order
θ	Instability threshold (upper limit) for each accepted rescheduling plan
$P_i/D_i/L_i$	The selling price/due date/deadline of order i
TC_i, TC_i'	Total production cost of order i
RMC_i/w_i	The raw material/unit uncertain penalty cost of job i
ϖ_i	The total weighted tardiness penalty of order i
$\mathrm{TPC}_{ij}/\mathrm{TPC}_i$	Total processing cost of operation O_{ij}/job i ($\mathrm{TPC}_i = \sum_{j=1}^{n_i} \mathrm{TPC}_{ij}$)
TUT_{ij}	Total unavailable time in processing operation O_{ij}
$X_l(t)/\widetilde{X}_l(t)$	The degradation process/degradation increment in the lth maintenance cycle
Z_l	The degradation level after the lth maintenance activity
α, β	The parameters related to a homogeneous Gamma process
a/L	The accelerated degradation factor/degradation failure threshold
p, q	The parameters related to a Beta distribution
$T_{\min}$	The allowed minimum inspection intervals ($T_{\min} > 0$)
T_s	The sth inspection interval in each maintenance cycle
m_1, m_2, m_3	The parameters related to preventive maintenance duration
C_1, C_2, C_3	The parameters related to operating cost
C_I/C_f	The cost of each inspection/downtime cost per unit time
$C_{\mathrm{PR}}/C_{\mathrm{CR}}/T_{\mathrm{RE}}$	The preventive replacement cost/corrective replacement cost/replacement time
$E_g(K_m)$	Expected inspection times with PM or PR in the mth maintenance cycle
$E_g(\mathrm{MI}_m)$	Expected maintenance interval with PM or PR in the mth maintenance cycle
$E_f(K_m)$	Expected inspection times with CR in the mth maintenance cycle
$E_f(\mathrm{MI}_m)$	Expected maintenance interval with CR in the mth maintenance cycle
$E(\mathrm{MT}_m)$	Expected maintenance duration with PM in the mth maintenance action
$E(\mathrm{FTR}_m)$	Expected waiting time from failure to inspection or replacement in the mth cycle
C_{PM}^0	Imperfect PM cost incurred when the deterioration level is reduced to 0
$E(C_{\mathrm{PM}}^{(m)})$	Expected PM cost in the mth maintenance action
$E(C_o^{(m)})$	Expected operating cost in the mth maintenance cycle
$E(W_f^{(m)})$	Expected waiting time from failure to the replacement in mth maintenance cycle
P_{gm}/P_{fm}	Probability that machine is performed PM or PR/CR at the end of the mth cycle
$E(\mathrm{MI}_l)/E(\mathrm{MD}_l)$	Expected maintenance interval/duration in the lth maintenance cycle
$C(N, T_{\max}, \xi, b_o)$	Expected average cost rate (ACR). For machine M_k, it is recorded as ACR_k
Decision variables	
N	Total number of maintenance intervals in the whole life cycle
$T_{\max}$	Maximum inspection interval in each maintenance cycle
ξ	Preventive maintenance threshold
b_o	The parameter related to the oth inspection policy
Y_i	If Y_i=1, then order i is accepted; otherwise, Y_i=0
Z_{ijk}	If Z_{ijk}=1, then O_{ij} is processed on machine M_k; otherwise, Z_{ijk}=0
U_{ijk}	If U_{ijk}=1, then O_{ij} is interrupted by maintenance on M_k; otherwise, U_{ijk}=0
V_{lij}	If $V_{lij} = 1$, then the lth maintenance is performed during the processing of operation O_{ij}; otherwise, V_{lij}=0
W_{ij}	The maximum deviation degree of O_{ij} compared to the previous scheduling plan

from customers, and the dispatcher needs to decide which orders to be accepted or rejected based on the real-time production status and limited production capacity. If some new orders are accepted, they must be scheduled together with the existing orders that are accepted in previous but some operations are not finished, i.e., rescheduling. Each incoming order i is identified with a release time R_i, raw material cost RMC_i, total processing cost TPC_i, selling price P_i, due date D_i, deadline L_i and n_i consecutive operations $O_i = \{O_{i1}, O_{i2}, \ldots, O_{in_i}\}$, and each operation O_{ij} can be processed by a set of alternative machines. If the manufacturer finishes a new inserted order i before the due date

D_i, the total production cost is $\mathrm{TC}_i = \mathrm{RMC}_i + \mathrm{TPC}_i$; otherwise, it will consume more cost (TC_i') that is equal to TC_i plus the tardiness penalty, i.e., $\mathrm{TC}_i' = \mathrm{TC}_i + \varpi_i$, where $\varpi_i = \max\{w_i(C_i^{\mathrm{com}} - D_i), 0\}$ and $w_i = |P_i - \mathrm{RMC}_i - \mathrm{TPC}_i| / (L_i - D_i)$. Note that, if the optimal scheduling plan becomes worse after a new order i is inserted, it should be rejected, and the firm needs to pay a certain amount of rejection cost, i.e., λP_i. Furthermore, the expected maintenance interval $[E(\mathrm{MI}_l)]$ and maintenance duration $[E(\mathrm{MD}_l)]$ of each machine is determined by the Gamma process with accelerated degradation factor, and the influence of many related factors is incorporated when making maintenance decisions. After that, the total processing cost of each machine is evenly distributed to each processed operation for calculating the total processing cost of each accepted order.

Based on the definition, the concerned problem in this study can be decomposed into the following five sub-problems: (1) decide whether to accept a new arrived order or not, i.e., order acceptance problem; (2) (re)assign each operation of accepted order(s) to a suitable machine from the candidate machine set, i.e., machine selection (MS) problem; (3) arrange the processing sequence of operations on the production machine, i.e., operation sequence (OS) problem; (4) select an appropriate rescheduling strategy from an adaptive rescheduling strategy pool (see Appendix A) according to different rescheduling scenarios to self-arrange all accepted orders, i.e., the selection problem of adaptive rescheduling strategies; and (5) determine the CBPM planning according to the processing time and degradation level, i.e., maintenance optimization problem.

3.3. CBPM policy

Let the random process $\{X(t), t \geq 0\}$ be the Gamma process, and the probability density function (pdf) in the lth maintenance cycle can be defined as (Cheng et al., 2018):

$$f_t^{(l)}(x) = \frac{\beta^{\alpha a^{l-1}t} x^{\alpha a^{l-1}t-1}}{\Gamma(\alpha a^{l-1}t)} e^{-\beta x}, \quad x, t \geq 0, \tag{1}$$

where $\alpha a^{l-1}t$ and β are the shape and scale parameters, and a is the accelerated degradation factor. The mean and variance of the degradation speed are $\alpha a^{l-1}/\beta$ and $\alpha a^{l-1}/\beta^2$, respectively. $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1}e^{-x}\mathrm{d}x$ is the Gamma function for $\alpha > 0$, and $\Gamma(\alpha, \epsilon) = \int_\epsilon^\infty x^{\alpha-1}e^{-x}\mathrm{d}x$ is the incomplete Gamma function for $\alpha > 0$ and $\epsilon \geq 0$.

Let $X_l(t)$ denote the deterioration state at time t in the lth maintenance cycle, and T_s denotes the inspection interval between the $(s-1)$-th and the sth inspections in each maintenance cycle. In the past, Grall et al. (2002) demonstrated that linear decreasing inspection interval seems to be an efficient policy. Based on this, we define a periodic and three non-periodic inspection policies as follows: (1) constant function: $T_s = \max\{T_{\max}, T_{\min}\}$; (2) linear decreasing function: $T_s = \max\{[1 - b_2(s-1)]T_{\max}, T_{\min}\}$; (3) convex decreasing function: $T_s = \max\{b_3^{s-1}T_{\max}, T_{\min}\}$; and (4) concave decreasing function: $T_s = \max\left\{\frac{\ln[b_4-(s-1)]}{\ln(b_4)}T_{\max}, T_{\min}\right\}$ if $b_4-(s-1) > 0$; otherwise, $T_s = T_{\min}$, where $0 < b_o < 1$ for $o = 2, 3$, and $b_4 > 1$.

According to the independent incremental property, the sum of $r-1$ independent increments in the lth maintenance cycle can be expressed as $Y_l = \widetilde{X}_l(T_1) + \widetilde{X}_l(T_2) + \cdots + \widetilde{X}_l(T_{r-1})$, which is also an independent increment from time 0 to time $\sum_{s=1}^{r-1} T_s$ in the lth maintenance cycle. Thus, the pdf of Y_l can be formulated as:

$$g_t^{(l)}(y_l) = \frac{\beta^{\alpha a^{l-1}t} y_l^{\alpha a^{l-1}t-1}}{\Gamma(\alpha a^{l-1}t)} e^{-\beta y_l}, \quad y_l, t \geq 0, \tag{2}$$

where $t = \sum_{s=1}^{r-1} T_s$.

In reality, PM activity is usually imperfect, i.e., PM restores the system back to an intermediate state between as-good-as new and as-bad-as right before maintenance (Rafiee et al., 2015). To depict the degradation level of right after the lth maintenance (i.e., residual degradation, denoted as Z_l), Beta distribution is widely employed (see blue dotted line in Fig. 1), and the pdf of right after the lth maintenance is formulated as (Pei et al., 2018):

$$h_{Z_l}^{(l)}(z_l) = \frac{1}{\xi}\frac{\Gamma(p^l+q)}{\Gamma(p^l)\Gamma(q)}\left(\frac{z_l}{\xi}\right)^{p^l-1}\left(1-\frac{z_l}{\xi}\right)^{q-1}, \quad 0 \leq z_l \leq \xi, \tag{3}$$

where $p, q > 1$ are parameters of Beta distribution.

According to the study of Huynh (2020b), the PM time (denoted as MT_l) of the lth maintenance is relevant to both the current state $X_l(t)$ and the post-maintenance state Z_l, i.e., $\mathrm{MT}_l = m_1 + m_2 X_l(t) + m_3 Z_l$. Meanwhile, to model the impact of imperfect maintenance action on the performance, the degradation improvement factor has been introduced by Do and Berenguer (2012), which is defined as the ratio of the improvement gain divided by the deterioration level of machine before maintenance [i.e., $(X_l(t) - Z_l)/X_l(t)$]. Based on the improvement factor, imperfect maintenance cost can be evaluated and considered as a function of the improvement factor, which can be defined as (Do et al., 2015): $C_{\mathrm{PM}}^{(l)} = C_{\mathrm{PM}}^0[1 - Z_l/X_l(t)]^\eta$, where η is a non-negative real number.

In the newly proposed maintenance policy, $C(t)$ is the total cost incurred in the time interval $[0, t]$. According to the renewal reward theorem (Ross, 1996), we have

$$C(N, T_{\max}, \xi, b_o) = \lim_{t\to\infty}\frac{C(t)}{t} = \frac{E(C)}{E(t)}, \tag{4}$$

where $E(C)$ and $E(t)$ are the expected total cost and the expected length of a renewal cycle.

According to the definition of decision variables, the machine is replaced at the Nth maintenance or after a failure, so the analytical expression of the expected total cost and expected length can be formulated as:

$$\begin{aligned} E(C) = {} & \left[C_I\sum_{l=1}^{N}E_g(K_l) + \sum_{l=1}^{N-1}E(C_{\mathrm{PM}}^{(l)}) + \sum_{l=1}^{N}E(C_o^{(l)}) + C_{\mathrm{PR}}\right]P_{g_N} + \\ & \sum_{m=1}^{N}\left[C_I\left(\sum_{l=1}^{m-1}E_g(K_l) + E_f(K_m)\right) + \sum_{l=1}^{m-1}E(C_{\mathrm{PM}}^{(l)}) + C_f E(W_f^{(m)}) \right. \\ & \left. + \sum_{l=1}^{m}E(C_o^{(l)}) + C_{\mathrm{CR}}\right]P_{f_m}, \end{aligned} \tag{5}$$

$$\begin{aligned} E(t) = {} & \left[\sum_{l=1}^{N}E_g(\mathrm{MI}_l) + \sum_{l=1}^{N-1}E(\mathrm{MT}_l) + T_{\mathrm{RE}}\right]P_{g_N} \\ & + \sum_{m=1}^{N}\left[\left(\sum_{l=1}^{m-1}E_g(\mathrm{MI}_l) + E_f(\mathrm{MI}_m)\right) + \sum_{l=1}^{m-1}E(\mathrm{MT}_l) + T_{\mathrm{RE}}\right]P_{f_m} \\ = {} & \left\{P_{g_N}\sum_{l=1}^{N}E_g(\mathrm{MI}_l) + \sum_{m=2}^{N}P_{f_m}\left[E_g(\mathrm{MI}_1) + E_g(\mathrm{MI}_2) + \cdots + E_g(\mathrm{MI}_{m-1})\right]\right. \\ & \left. + \sum_{m=1}^{N}P_{f_m}E_f(\mathrm{MI}_m)\right\} + \left\{P_{g_N}\left[E(\mathrm{MT}_1) + E(\mathrm{MT}_2) + \cdots + E(\mathrm{MT}_{N-1})\right]\right. \\ & \left. + \sum_{m=2}^{N}P_{f_m}\left[E(\mathrm{MT}_1) + E(\mathrm{MT}_2) + \cdots + E(\mathrm{MT}_{m-1})\right]\right\} \\ & + \left(P_{f_1} + P_{f_2} + \cdots + P_{f_N} + P_{g_N}\right)T_{\mathrm{RE}} \\ = {} & \sum_{l=1}^{N}\left[P_{g_l}E_g(\mathrm{MI}_l) + P_{f_l}E_f(\mathrm{MI}_l)\right] + \sum_{l=1}^{N-1}\left[P_{g_l}E(\mathrm{MT}_l) + P_{f_l}T_{\mathrm{RE}}\right] \\ & + \left(P_{g_N} + P_{f_N}\right)T_{\mathrm{RE}}, \end{aligned} \tag{6}$$

where $P_{g_N} + P_{f_m} + P_{f_{m+1}} + \cdots + P_{f_N} = 1 - P_{f_1} - P_{f_2} - \cdots - P_{f_{m-1}} = P_{g_{m-1}}$. Meanwhile, the means and calculation processes of P_{g_m}, P_{f_m}, $E_g(K_m)$, $E_g(\mathrm{MI}_m)$, $E_f(K_m)$, $E_f(\mathrm{MI}_m)$, $E(\mathrm{MT}_m)$, $E(C_{\mathrm{PM}}^{(m)})$, $E(W_f^{(m)})$ and $E(C_o^{(m)})$ are given in Appendix B, and the expected maintenance interval [i.e., $E(\mathrm{MI}_l)$] and expected maintenance duration [i.e., $E(\mathrm{MD}_l)$] are calculated as:

$$E(\mathrm{MI}_l) = P_{g_l}E_g(\mathrm{MI}_l) + P_{f_l}E_f(\mathrm{MI}_l), \tag{7}$$

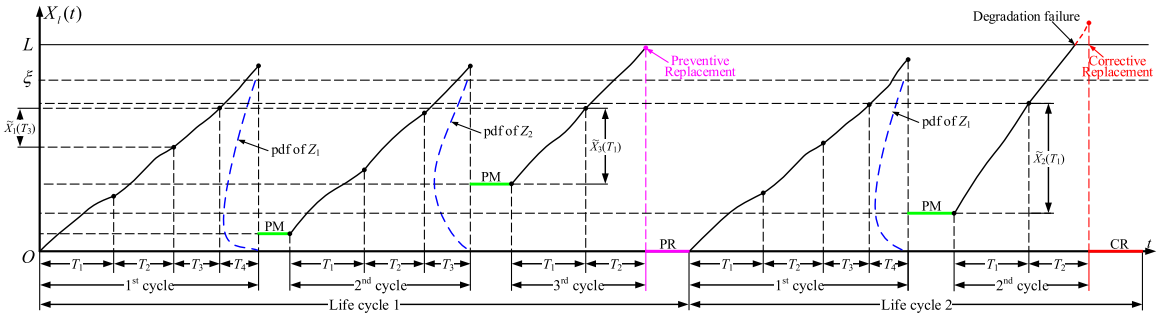


Fig. 1. Illustration of a degradation trajectory and impacts of imperfect maintenance ($N = 3$).

$$E(\mathrm{MD}_l) = \begin{cases} P_{g_l} E(\mathrm{MT}_l) + P_{f_l} T_{\mathrm{RE}}, & l = 1, 2, \ldots, N-1, \\ \left(P_{g_N} + P_{f_N}\right) T_{\mathrm{RE}}, & l = N. \end{cases} \tag{8}$$

Therefore, the expected total unavailable time in the lth maintenance cycle is the sum of $E(\mathrm{MD}_l)$ and $E(\mathrm{FTR}_l)$, where $E(\mathrm{FTR}_l) = P_{f_l} E(W_f^{(l)})$ is the average waiting time from failure to replacement (i.e., downtime).

When the optimal maintenance decisions are determined, the expected total cost should be equally allocated to the corresponding processed operations to calculate the total processing cost of each order, i.e., $\mathrm{TPC}_i = \sum_{j=1}^{n_i} \mathrm{TPC}_{ij}$, where $\mathrm{TPC}_{ij} = \sum_{k=1}^{K} Z_{ijk} P_{ijk} \mathrm{PC}_k$ is the total processing cost of operation O_{ij}, in which $\mathrm{PC}_k = E(C)/E(\mathrm{PT})$ is the average processing cost rate of machine M_k, and $E(\mathrm{PT}) = \sum_{l=1}^{N} \left[E(\mathrm{MI}_l) - E(\mathrm{FTR}_l)\right]$ is the total working time.

3.4. Multi-objective optimization model

In this work, a multi-objective optimization model is developed, which aims to simultaneously minimize the makespan f_1, scheduling instability f_2 and cumulative production cost f_3 at each rescheduling moment. The mathematical model can be formulated as:

$$\text{Min} \quad \mathbf{F} = (f_1, f_2, f_3), \tag{9}$$

where

$$f_1 = \max\{C_i^{\mathrm{com}} \mid i = 1, 2, \ldots, N_A + N_N\}, \tag{10}$$

$$f_2 = \sum_{i=1}^{N'_A} \sum_{j=1}^{n'_i} W_{ij} / Q, \tag{11}$$

where W_{ij} represents the maximum deviation degree compared to the previous scheduling plan, and defined as (Zakaria & Petrovic, 2012):

$$W_{ij} = \begin{cases} 1, & \text{if the processing machine of } O_{ij} \text{ changes,} \\ 0.6, & \text{if the processing sequence of } O_{ij} \text{ changes,} \\ 0.3, & \text{if the start time of } O_{ij} \text{ changes,} \\ 0, & \text{if there is no change,} \end{cases}$$

$$f_3 = \sum_{i=1}^{N_A + N_N + N_R} \mathrm{TC}_i, \tag{12}$$

where TC_i is the total production cost of order i, and calculated as $\mathrm{TC}_i = \left[\mathrm{RMC}_i + \mathrm{TPC}_i + \varpi\right] Y_i + \lambda(1 - Y_i) P_i$. Subject to:

$$\max\{E_{i'j'k}, E_{i(j-1)k'}\} \le S_{ijk}, \quad \forall i, i', j, j', k, k', \tag{13}$$

$$S_{ijk} + \left[(P_{ijk} + \mathrm{TUT}_{ij}) U_{ijk} + P_{ijk}(1 - U_{ijk})\right] Z_{ijk} \le E_{ijk}, \quad \forall i, j, k, \tag{14}$$

$$\mathrm{TUT}_{ij} = \sum_{l=1}^{N_k} [E(\mathrm{FTR}_l) + E(\mathrm{MD}_l)] V_{lij}, \quad \forall i, j, \tag{15}$$

$$\sum_{k=1}^{m} Z_{ijk} = 1, \quad \forall i, j, \tag{16}$$

$$\sum_{i=1}^{n} \sum_{j=1}^{n_i} Z_{ijk} = 1, \quad \forall k, \tag{17}$$

$$Y_i \in \{0,1\}, Z_{ijk} \in \{0,1\}, U_{ijk} \in \{0,1\}, V_{lij} \in \{0,1\} \quad \forall i, j, k, l. \tag{18}$$

In the mathematical model, Expressions (10)–(12) are the objective functions. Constraint (13) is the operation precedence constraint, which means that an operation must start to process after its preceding operations are completed. Constraint (14) implies that the start processing time of O_{ij} should be less than or equal to the end processing time. If there exists a maintenance activity ($U_{ijk} = 1$), the time occupied by this operation is $P_{ijk} + \mathrm{TUT}_{ij}$; otherwise, it is P_{ijk}. Constraint (15) is used to calculate the total unavailable time (TUT_{ij}) in processing O_{ij}. Note that, the total unavailable time is the sum of all unavailable durations that interrupt the operation. Constraint (16) indicates that one operation can only be processed by one machine at a time. Constraint (17) reveals that one machine can only be allocated to deal with only one task at the same time. Constraint (18) shows that Y_i, Z_{ijk}, U_{ijk} and V_{lij} are binary decision variables.

3.5. Illustrative example

To clearly illustrate the concerned problem and three adopted rescheduling strategies (denoted as S1, S2 and S3, see Appendix A and An, Chen, Gao, Li, & Zhang, 2022), Instance 3 in Appendix H is employed, in which jobs 1–10, 11, 12 and 13 arrive at time "0", "15", "25" and "35" (assume that the selected rescheduling method sequence is S2, S1 and S3), respectively. The initial scheduling is shown in Fig. 2 and the objective values are $f_1 = 62.8540$, $f_2 = 0$ and $f_3 = 1.5105\mathrm{E}+05$ (the calculation process is shown in Appendix C1). At time "15", S2 is applied to rescheduling, and the target values are $f_1 = 70.7918$, $f_2 = 0.2118$ and $f_3 = 1.6758\mathrm{E}+05$ (the calculation process is described in Appendix C2). Applying the TOPSIS method (Tavana et al., 2016) to evaluate the initial scheduling plan with rejected order 11 and the current rescheduling plan, it is found that the scheduling performance of rejecting order 11 is better, so the target values of the initial scheduling plan become $f_1 = 62.8540$, $f_2 = 0$ and $f_3 = 3.5951\mathrm{E}+05$. At time "25", S1 is used for rescheduling, and the desired values are $f_1 = 113.2282$, $f_2 = 0$ and $f_3 = 3.8005\mathrm{E}+05$ (the calculation process is explained in Appendix C3). Similarly, the TOPSIS evaluation results also show that the current solution is still inferior to the initial solution with rejected orders 11 and 12, so the desired values of the initial solution become $f_1 = 62.8540$, $f_2 = 0$ and $f_3 = 6.2299\mathrm{E}+05$. At time "35", S3 is employed to rescheduling, and the goal values are $f_1 = 99.0060$, $f_2 = 0$ and $f_3 = 6.4063\mathrm{E}+05$ (the calculation process is displayed in Appendix C4). Using the same evaluation method, the results show that rejecting orders 11–13 is better, so the goal values of the initial scheduling plan become $f_1 = 62.8540$, $f_2 = 0$ and $f_3 = 8.9067\mathrm{E}+05$. All in all, accepted orders are 1–10, rejected orders are 11–13, and the optimal scheduling schemes at different rescheduling moments are the same as sketched in Fig. 2, but the cumulative production cost is unequal since some orders are rejected in rescheduling point.

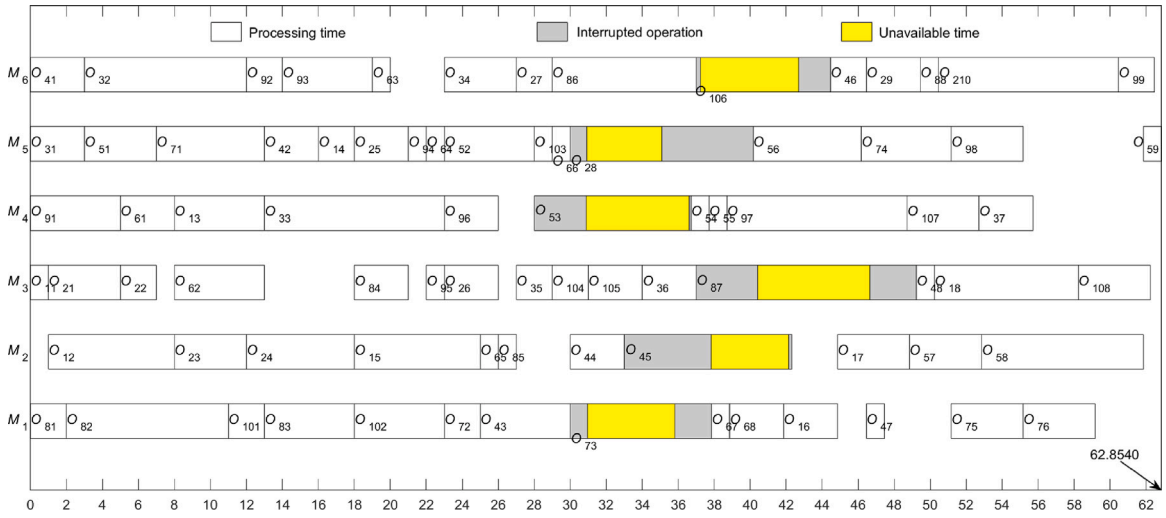


Fig. 2. Optimal scheduling scheme in all (re)scheduling points for Instance 3.

4. Proposed algorithm: HMOEA

In this section, the framework of the proposed HMOEA algorithm is first presented. Second, the chromosome representation and population initialization method are designed. Next, the genetic operators and a hybrid local search with N5 structure, processing time window and orthogonal learning are constructed.

4.1. Framework of HMOEA

The main procedure of the proposed algorithm is displayed in Algorithm 1, which is explained from the following two phases. In Phase 1 ($c=1$): (1) a reference point set Z^r and initial acceptance jobset (i.e., $AcceptedSet$) are generated (lines 1–2); (2) a set of offspring population P_g is produced by genetic operators and hybrid LS mechanism under any given jobset (lines 4–28). Note that, if the distribution metric (i.e., $AverageDeviation$) of the solutions exceeds the predetermined threshold (i.e., $\lceil N \times f_r \rceil$), RV is updated (lines 24–27); and (3) an optimal solution is selected after inserting each new order, and then update the temporary optimal solution (i.e., $Bestsolution$) and order acceptance jobset (i.e., $AcceptedSet$), as shown in lines 29–34.

In Phase 2 ($c > 1$): (1) acquire the newly inserted order information, and update $AcceptedSet$, $CurrentSeq$ and optimal solution (i.e., $Bestsolution$) from the last rescheduling stage (lines 38–41); (2) according to the current situation, we adaptively choose an appropriate rescheduling strategy to perform rescheduling (lines 43–44); (3) three rescheduling strategies can be selectively executed, but the jobs involved in different rescheduling strategies are distinct (lines 45–59), e.g., real-time rescheduling strategy (S1) only concerns the new inserted jobs; (4) after the rescheduling, the impact of insertion new job needs to be evaluated, and $AcceptedSet$ and $Bestsolution$ should be updated (line 60); and (5) after each round of rescheduling, we need to update the current optimal solution (i.e., $Bestsolution$) to facilitate production execution and target value calculation in the next rescheduling stage (line 62). It is worth noting that, if the order selection and acceptance only occurs at the initial, only Phase 1 is executed; otherwise (i.e., $Cycle > 1$), Phase 2 needs to be executed to deal with the dynamic events.

4.2. Chromosome representation and population initialization

In the proposed algorithm, each chromosome contains two vectors: operation sequence (OS) and machine selection (MS). Each dimension in OS is represented by the job number, and the occurrence times of the job number is the operation number, e.g., "2" appearing for the third time means the third operation of Job 2, namely O_{23}. Whereas each element in MS is the machine name, e.g., the MS gene corresponding to O_{23} is "5", which means that the operation is processed on machine M_5. For each chromosome, the active decoding method (Pinedo, 2002) is employed to decline the makespan and machine idle time. For easy understanding, a partial FJSP instance is given in Appendix D (assuming Jobs 1, 2 and 4 are accepted). A feasible solution representation is reported in Fig. 3, and the decoding result with unavailable duration is displayed in Fig. 4 (O_{13} and O_{42} are interrupted by maintenance activities).

Generally, the quality of the initial population may significantly affect the convergence speed of the algorithm to find satisfactory solutions in a reasonable computational time (Gao, Yang, et al., 2019). Therefore, generating a high-quality initial population plays a critical role in the algorithm iteration. A single initialization method cannot effectively guarantee the population quality, so the hybrid initialization method has become one of the popular approaches (An, Chen, Gao, Li, & Zhang, 2022; Pezzella et al., 2008; Zhang et al., 2011). In this work, three rules (named GCC) are employed, which are related to the total workload, completion time and processing cost, respectively. The detailed procedures of the hybrid initialization method are described as follows: (1) global minimum-processing time (G): select the machine (denoted as M_G) with the minimum total processing time in the candidate machine set AMS_{ij}; (2) minimum-completion time (C): select the machine (named as M_C) with the earliest completion time on all selectable machines; and (3) minimum-processing cost (C): calculate the average processing cost of operation O_{ij} on machines M_G and M_C, and denoted as PC_G and PC_C. If $PC_G \leq PC_C$, M_G is selected for processing O_{ij}; otherwise, M_C is selected. Note that, after the processing machine of O_{ij} is assigned, the machine workload and resource (e.g., machine and job) available time need to be updated.

4.3. Genetic operators

The essence of genetic operators (including crossover and mutation) is to produce a small perturbation to individuals, thereby achieving a search for the whole feasible space (Zhang et al., 2007). For OS crossover, if the chromosome (e.g., P_1) contains multiple jobs, a modified precedence operation crossover 2 (MPOX2) (An, Chen, Gao, Zhang, et al., 2022) is utilized for the individual crossover; otherwise, there is no crossover for OS since there is only one job. More precisely, MPOX2 is shown in Fig. 5, when the individuals P_1 and P_2 are selected from the tournament pool by TOPSIS method (Tavana et al., 2016), the crossover can be divided into the following three steps: (1) the job classification is generated twice (i.e., JS_1 and JS_2; and JS'_1 and JS'_2); (2) the jobsets

Algorithm 1 Main procedure of HMOEA

Input: $Cycle$, G, N, P_c, P_m, f_r, γ_0, γ_c, λ, θ, etc.
1: Z^r=GenerateReferencePoints(); Sort the newly inserted job(s), and the sequence is denoted as $CurrentSeq$;
2: $AcceptedSet = CurrentSeq(1:\lfloor \gamma_0 \times |CurrentSeq| \rfloor)$; $CurrentSeq = CurrentSeq \backslash AcceptedSet$; $Bestsolution = \varnothing$;
3: **for** $Jobnum = CurrentSeq$ **do**
4: P_1=Initialization(N, $Jobnum \bigcup AcceptedSet$); %$N$ is the population size.
5: **for** $g = 1 : G$ **do** %G is the maximum number of iterations.
6: P_{Cro}=Crossover(P_g,P_c); P_{Mut}=Mutation(P_g,P_m); P_{LS}=LocalSearch($P_{Cro} \bigcup P_{Mut}$); U_g=$P_g \bigcup P_{LS}$;
7: (F_1,F_2,...)=NonDominatedSorting(U_g); let $P_{g+1} = S_g = \varnothing$ and $k = 1$;
8: **if** $|U_g| \leq N$ **then**
9: P_{g+1}=U_g, return;
10: **else**
11: **repeat**
12: S_g=$S_g \bigcup F_k$ and $k = k + 1$;
13: **until** $|S_g| \geq N$
14: **if** $|S_g| = N$ **then**
15: P_{g+1}=S_g, return;
16: **else**
17: P_{t+1}=$\bigcup_{p=1}^{p=k-2} F_p$; The critical layer $F_l = F_{k-1}$;
18: **while** $|P_{g+1}| < N$ **do**
19: P_{g+1}=$P_{g+1} \bigcup F_l(r,:)$; %Select one individual from F_l by niche-preservation operator (Deb & Jain, 2014).
20: F_l=$F_l \backslash F_l(r,:)$; %Remove the selected individual from F_l.
21: **end while**
22: **end if**
23: **end if**
24: rho=AssociateToReferencePoint(P_{g+1},Z^r); $AverageDeviation$=sum(| rho–mean(rho)|)/N;
25: **if** mod(g,$\lceil G \times f_r \rceil$)==0 && $AverageDeviation \geq \lceil N \times f_r \rceil$ **then**
26: Z^r=AdaptiveReferencePoints(Z^r,P_{g+1}); %Update the reference vectors (Cheng et al., 2016).
27: **end if**
28: **end for**
29: $Temp = P_{G+1}$(The instability of the production schemes is less than the predefined instability threshold θ);
30: $Temp = Bestsolution \bigcup Temp$; %Combine the populations.
31: **if** $Temp$ is a non-empty set **then**
32: Chose a $Bestsolution$ from $Temp$ according to the TOPSIS method (Tavana et al., 2016);
33: Update $AcceptedSet$ according to $Bestsolution$;
34: **end if**
35: **end for**
36: Let $Input = Bestsolution$;
37: **for** $c = 2 : Cycle$ **do** %$Cycle$ is total scheduling times.
38: Acquire new arrived order information, e.g., quantity, processing machines/time, due dates and deadlines;
39: Sort the newly inserted job(s), and the sequence is denoted as $CurrentSeq$;
40: Update $AcceptedSet = CurrentSeq(1:\lfloor \gamma_c \times |CurrentSeq| \rfloor)$; $CurrentSeq = CurrentSeq \backslash AcceptedSet$;
41: Update the optimal solution (i.e., $Bestsolution$) from the last stage, i.e., reject all current inserted orders;
42: **for** $Jobnum = CurrentSeq$ **do**
43: Look for unfinished jobs and their unstarted operations in the previous scheduling plan, named $UnfinishedJobSet$;
44: Choose an appropriate rescheduling strategy (RS) according to the current rescheduling situation;
45: **switch** RS **do**
46: **case** $RS = 1$ %Real-time rescheduling strategy (S1).
47: Update machine and maintenance informations, and inherit the target from the previous scheduling plan;
48: P_1=Initialization(N, $Jobnum \bigcup AcceptedSet$), and perform lines 5–28;
49: **end case**
50: **case** $RS = 2$ %Complete rescheduling strategy (S2).
51: Update the earliest available time of the machine and job, and maintenance informations;
52: The operations not involved in rescheduling keep the same as the original scheduling plan;
53: P_1=Initialization(N, $Jobnum \bigcup UnfinishedJobSet \bigcup AcceptedSet$, $Input$), and perform lines 5–28;
54: **end case**
55: **case** $RS = 3$ %Continuous-time insert rescheduling strategy (S3).
56: P_1=Initialization(N, $Jobnum \bigcup AcceptedSet$), and perform lines 5–28;
57: Insert the individuals from P_G into the $Input$ in turn, and select N best individuals recorded as P_{G+1};
58: **end case**
59: **end switch**
60: Perform lines 29–34;
61: **end for**
62: Let $Input = Bestsolution$;
63: **end for**
Output: Output all optimal solutions (i.e., $Bestsolution$) and all accepted orders (i.e., $AcceptedSet$).

with a large number of jobs are inherited to the corresponding child [i.e., $P_1(JS_1) \rightarrow C_1(JS_1)$ and $P_2(JS_1') \rightarrow C_2(JS_1')$]; and (3) the left genes in C_1 and C_2 of OS string are migrated from two parents and maintain their encoding sequence, i.e., $P_2(JS_2) \rightarrow C_1(JS_2)$ and $P_1(JS_2') \rightarrow C_2(JS_2')$. For MS crossover, multi-point crossover (Wang et al., 2010) is adopted. For individual mutation, a random whole neighborhood (Gao et al.,

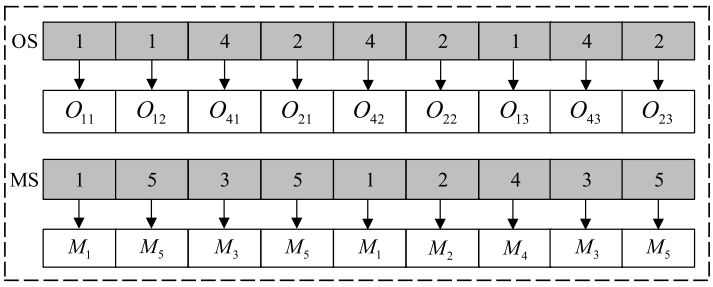


Fig. 3. Chromosome representation of a feasible solution.

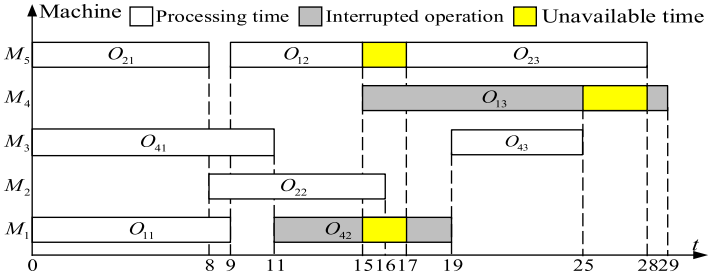


Fig. 4. Scheduling plan with maintenance for a feasible solution.

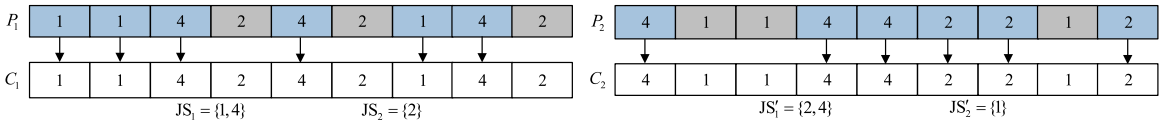


Fig. 5. MPOX2 for the OS crossover.

2015) is employed for OS, and the roulette wheel method (RWM, short processing time first or small average processing cost first) is adopted to select a new machine from the available machine set for each processed operation in MS. Note that, the above crossover and mutation may cause the MS gene to be infeasible, thus RWM is utilized to correct the gene.

4.4. Hybrid local search with N5 structure, processing time window and orthogonal learning

To acquire a desirable solution, it is common to integrate a local search method within a standard algorithm (Li et al., 2019). Consequently, there are amounts of LS methods are proposed, e.g., N5 structure (Nowicki & Smutnicki, 1996) and operation migration (An, Chen, Gao, Li, & Zhang, 2022). According to previous research findings (An, Chen, Gao, Zhang, et al., 2022), the operation migration based on the processing time window and N5 structure can effectively improve the performance of the algorithm. However, they mainly focus on the optimization of the processing sequence (i.e., OS) and lack of effective exploration for machine selection. For this reason, OL strategy is utilized to quickly find the best combination of the processing machines for several randomly selected operations. To conclude, the execution procedure of the hybrid LS is described as follows: (1) find the critical path of the chromosome, and determine the latest start time [i.e., $L^s(O_{ij})$] and earliest completion time [i.e., $E^c(O_{ij})$] of each operation O_{ij}; (2) use the idle-oriented neighborhood structure (i.e., change the processing machine or sequence of critical operations based on processing time window) to shorten the critical path and output an optimal solution; (3) employ N5 structure to change the processing sequence on the obtained solution of Step (2), and then output Pareto-optimal solutions; and (4) randomly select several operations (each selected operation has at least two optional machines) of a solution that comes from Step (3) to generate an orthogonal combination of optional machines, and then use factor analysis to determine the best combination of these machines. For the sake of exposition, an example is depicted in Fig. 6, and the "fit" is denoted as response value (the larger is better). Through orthogonal experiment analysis (Xiong et al., 2014), it is found that the best processing machine array for the selected three operations is "4 4 3", which can greatly decline the computational burden and improve the search quality.

5. Performance analysis of the proposed CBPM policy

In this section, the parameter estimation methods are first introduced. Second, a discrete approximate iteration (DAI) algorithm is designed to find the optimal solution. Finally, a discussion about the proposed inspection and maintenance policies is provided.

5.1. Parameter estimation

To optimize the long-run expected cost rate, we should first estimate the pdf of the Gamma process $X_l(t)$ in the lth maintenance cycle, i.e., parameters α, β and a. Let $\alpha_l \Delta t_{ls}$ and β_l be the shape and scale parameters of $X_l(t)$, respectively. Let t_{ls} and x_{ls} denote the sth inspection time and the corresponding state value in the lth maintenance cycle for $l = 1, 2, \ldots, N$ and $s = 1, 2, \ldots, M$. Given the increments $\Delta t_{ls} = t_{ls} - t_{l(s-1)}$ with $\Delta t_{l1} = t_{l1}$ and $\Delta x_{ls} = x_{ls} - x_{l(s-1)}$ with $\Delta x_{l1} = x_{l1}$, and the pdf of degradation increment is $f^{(l)}_{\Delta t_{ls}}(\Delta x_{ls}) = \frac{\beta_l^{\alpha_l \Delta t_{ls}} \Delta x_{ls}^{\alpha_l \Delta t_{ls}-1}}{\Gamma(\alpha_l \Delta t_{ls})} e^{-\beta_l \Delta x_{ls}}$, so the log-likelihood function of the joint pdf (i.e., $\prod_{s=1}^{M} \frac{\beta_l^{\alpha_l \Delta t_{ls}} \Delta x_{ls}^{\alpha_l \Delta t_{ls}-1}}{\Gamma(\alpha_l \Delta t_{ls})} e^{-\beta_l \Delta x_{ls}}$) is $\ln L(\alpha_l, \beta_l) = \sum_{s=1}^{M} \alpha_l \Delta t_{ls} \ln \beta_l + (\alpha_l \Delta t_{ls} - 1) \ln \Delta x_{ls} - \beta_l \Delta x_{ls} - \ln \Gamma(\alpha_l \Delta t_{ls})$.

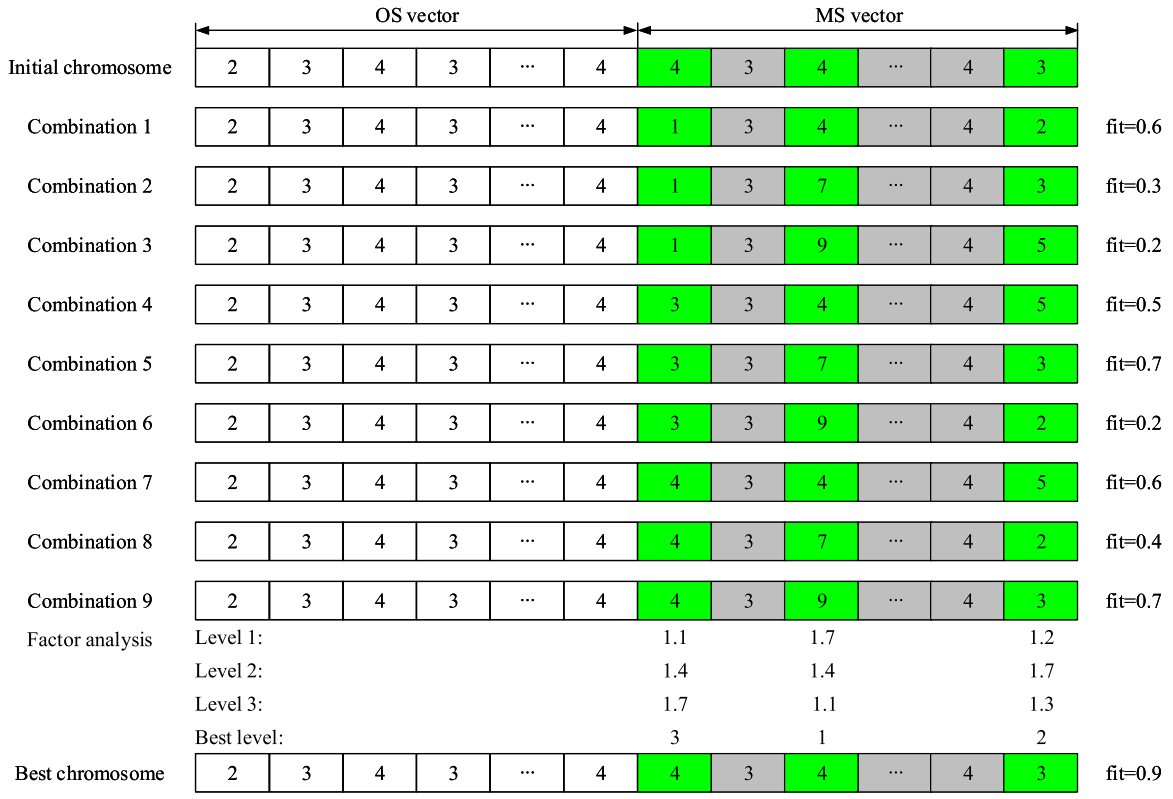


Fig. 6. Orthogonal experiment design (OED) for a non-dominated solution.

Differentiating $\ln L(\alpha_l, \beta_l)$ with respect to α_l and β_l, and setting them to zeros, i.e.,

$$\begin{cases} \frac{\partial \ln L(\alpha_l,\beta_l)}{\partial \alpha_l} = \sum_{s=1}^{M} \Delta t_{ls}[\ln \beta_l + \ln \Delta x_{ls} - \psi(\alpha_l \Delta t_{ls})] = 0, \\ \frac{\partial \ln L(\alpha_l,\beta_l)}{\partial \beta_l} = \sum_{s=1}^{M} \frac{\alpha_l \Delta t_{ls}}{\beta_l} - \Delta x_{ls} = 0, \end{cases} \tag{19}$$

where $\psi(\cdot) = \Gamma'(\cdot)/\Gamma(\cdot)$ is the digamma function.

Through solving Eq. (19), we can obtain the maximum likelihood estimators of α_l and β_l, denoted as $\hat{\alpha}_l$ and $\hat{\beta}_l$. Furthermore, since the mean deterioration rate $u_l = \hat{\alpha}_l/\hat{\beta}_l = a^{l-1}u_1$, where $u_1 = \hat{\alpha}_1/\hat{\beta}_1$, the parameter a can be obtained by using the least square method (Li et al., 2020).

For the parameters of Beta distribution (i.e., p_l and q_l), Liao et al. (2006) pointed out that the values can be estimated by the mean and variance of residual degradation (Z):

$$\begin{cases} E\left(\frac{Z}{\xi}\right) = \frac{p_l}{p_l+q_l} = 1 - \exp(-lu), \\ \mathrm{Var}\left(\frac{Z}{\xi}\right) = \frac{p_l q_l}{(p_l+q_l)^2+(p_l+q_l+1)} = \sigma^2, \end{cases} \tag{20}$$

where $u, \sigma^2 > 0$, which also can be obtained by using the maximum likelihood estimation (MLE) based on the residual degradation data (Liao et al., 2006; Pei et al., 2018). According to Eq. (3), there exists $\hat{p}_l = p^l$, so the parameter p can be obtained by data fitting. Similarly, the related parameters of imperfect PM time and operational cost can also be obtained by data fitting. Based on these, the used parameters can be determined as displayed in Appendix E.

5.2. Discrete approximate iteration (DAI) algorithm

The decision goal is to find the optimal mixed $(N^*, T^*_{\max}, \xi^*, b^*_o)$ policy to minimize the long-run average cost rate. To the best of our understanding, it is extremely difficult to derive the optimal solution $(N^*, T^*_{\max}, \xi^*, b^*_o)$ in mathematical theory (Huynh, 2020a). Thus, DAI algorithm is designed to find an optimal or a near-optimal solution, and the procedure is displayed in Appendix F and mainly consists of the following five steps: (1) input some non-optimized parameters and set the initial values of four decision variables (i.e., N, $T_{\max}$, ξ and b_o) and four recording functions [i.e., $C(N^*, T_{\max}, \xi, b_o)$, $C(N^*, T^*_{\max}, \xi, b_o)$, $C(N^*, T^*_{\max}, \xi^*, b_o)$ and $C(N^*, T^*_{\max}, \xi^*, b^*_o)$]; (2) find the optimal value $C(N^*, T_{\max}, \xi, b_o)$ under initial values of $T_{\max}$, ξ and b_o; (3) find the optimal value $C(N^*, T^*_{\max}, \xi, b_o)$ under fixed values of N^*, ξ and b_o; (4) find the optimal value $C(N^*, T^*_{\max}, \xi^*, b_o)$ under fixed values of N^*, $T^*_{\max}$ and b_o; and (5) find the optimal value $C(N^*, T^*_{\max}, \xi^*, b^*_o)$ under optimal values of N^*, $T^*_{\max}$ and ξ^*. Since the maintenance parameter setting ensures that the expected cost rate function is convex, the optimal solution of each independent decision variable can be found separately.

5.3. Discussion on the optimal inspection and maintenance policies

The DAI algorithm is utilized to seek the optimal maintenance decisions for each production machine, and the computational results are shown in Appendix G. According to the principle of minimum average cost rate, the optimal inspection policy (OIP) and its corresponding maintenance planning in each production machine can be obtained, as shown in Table 2. From Table 2, it can be observed that (1) the optimal inspection policies for different machines or maintenance parameter settings are variable, but in general, non-periodic concave decreasing inspection policy is a preferred choice for the homogeneous Gamma process; (2) the expected maintenance intervals [i.e., $E(\mathrm{MI}_l)$] are gradually shrinking due to the accelerated degradation and imperfect maintenance; and (3) the expected maintenance duration [i.e., $E(\mathrm{MD}_l)$] and expected waiting time from failure to replacement [i.e., $E(\mathrm{FTR}_l)$] are not strictly monotonic, since they are closely related to the improvement factor and degradation level. In addition, in order to verify that the DAI algorithm can find the optimal solution, taking Machine 2 and inspection Policy 4 as an example, the optimal solutions under four decision variables are sketched in Fig. 7 (convexity). As can be seen from Fig. 7, we can find that $T_{\max}$ and ξ are more sensitive than other two parameters.

Table 2
Optimal maintenance decisions under four inspection policies.

M_k	OIP	N^*	$T^*_{\max}$	ξ^*	b^*_o	ACR*	$E(\mathrm{MI}_l)$	$E(\mathrm{MD}_l)$	$E(\mathrm{FTR}_l)$
1	2	8	18.6	16.6	0.1	220.68	30.98; 20.86; 16.97; 13.60; 10.82; 8.59; 6.82; 5.40	8.08; 4.84; 4.09; 3.46; 2.94; 2.55; 2.25; 6.29	2.45E−02; 1.54E−02; 1.66E−02; 1.65E−02; 1.51E−02; 1.28E−02; 1.03E−02; 7.89E−03
2	4	12	18.7	17.3	8.6	212.17	33.83; 23.98; 21.53; 19.24; 17.14; 15.24; 13.54; 12.03; 10.69; 9.51; 8.46; 7.52	7.62; 4.30; 3.82; 3.38; 2.97; 2.60; 2.28; 2.00; 1.77; 1.58; 1.43; 6.38	6.57E−03; 3.22E−03; 3.16E−03; 3.05E−03; 2.88E−03; 2.66E−03; 2.40E−03; 2.13E−03; 1.85E−03; 1.59E−03; 1.35E−03; 1.13E−03
3	4	12	17.1	16.7	13.9	233.04	32.43; 23.91; 20.93; 17.99; 15.23; 12.77; 10.67; 8.92; 7.48; 6.24; 5.12; 4.05	9.78; 6.23; 5.43; 4.66; 3.96; 3.38; 2.93; 2.61; 2.41; 2.31; 2.27; 6.44	5.36E−03; 3.16E−03; 3.95E−03; 4.81E−03; 5.49E−03; 5.78E−03; 5.60E−03; 5.03E−03; 4.28E−03; 3.51E−03; 2.79E−03; 2.10E−03
4	4	8	13.5	16.7	12.1	247.68	28.91; 21.13; 18.62; 16.18; 13.84; 11.66; 9.67; 7.89	7.86; 5.70; 5.08; 4.44; 3.82; 3.23; 2.68; 6.83	1.84E−02; 6.52E−03; 5.94E−03; 5.29E−03; 4.63E−03; 4.02E−03; 3.49E−03; 3.05E−03
5	4	10	17.5	16.4	6.8	224.89	30.93; 21.22; 18.07; 15.16; 12.56; 10.31; 8.39; 6.74; 5.28; 3.97	7.34; 4.16; 3.60; 3.11; 2.70; 2.39; 2.16; 2.00; 1.88; 5.32	1.11E−02; 6.30E−03; 7.30E−03; 7.98E−03; 8.15E−03; 7.80E−03; 7.03E−03; 6.02E−03; 4.90E−03; 3.72E−03
6	1	13	17.4	18.0	–	216.80	34.24; 26.72; 24.53; 22.27; 20.01; 17.83; 15.80; 13.98; 12.40; 11.06; 9.93; 8.95; 8.07	8.07; 5.46; 4.94; 4.39; 3.84; 3.32; 2.84; 2.43; 2.11; 1.86; 1.71; 1.63; 6.97	1.95E−03; 9.56E−04; 1.08E−03; 1.24E−03; 1.42E−03; 1.57E−03; 1.66E−03; 1.64E−03; 1.53E−03; 1.35E−03; 1.15E−03; 9.55E−04; 7.90E−04
7	4	8	15.4	14.7	7.9	249.69	28.20; 20.87; 17.98; 15.21; 12.65; 10.38; 8.43; 6.82	8.00; 5.40; 4.75; 4.11; 3.54; 3.03; 2.62; 7.94	1.18E−02; 7.03E−03; 7.95E−03; 8.79E−03; 9.34E−03; 9.39E−03; 8.90E−03; 7.96E−03
8	3	10	19.6	18.1	0.8	223.40	31.74; 23.27; 20.71; 18.34; 16.18; 14.22; 12.47; 10.90; 9.50; 8.24	7.69; 4.67; 4.15; 3.68; 3.24; 2.86; 2.53; 2.24; 1.99; 6.48	1.14E−02; 6.01E−03; 5.82E−03; 5.52E−03; 5.14E−03; 4.70E−03; 4.22E−03; 3.73E−03; 3.25E−03; 2.79E−03
9	4	7	19.4	14.9	7.7	205.79	34.64; 23.54; 19.32; 15.45; 12.12; 9.37; 7.17	10.01; 5.87; 4.95; 4.15; 3.50; 3.01; 6.38	1.88E−02; 1.17E−02; 1.43E−02; 1.62E−02; 1.67E−02; 1.57E−02; 1.35E−02
10	2	12	18.7	16.1	0.1	254.97	32.29; 22.32; 18.99; 15.99; 13.36; 11.10; 9.20; 7.59; 6.22; 5.03; 3.98; 3.02	9.07; 5.31; 4.57; 3.91; 3.34; 2.87; 2.50; 2.21; 2.00; 1.84; 1.71; 5.38	1.54E−02; 8.47E−03; 8.69E−03; 8.56E−03; 8.07E−03; 7.28E−03; 6.31E−03; 5.29E−03; 4.32E−03; 3.43E−03; 2.62E−03; 1.90E−03
11	4	11	16.8	16.0	8.8	247.43	30.59; 22.85; 19.90; 17.03; 14.38; 12.04; 10.06; 8.43; 7.09; 5.94; 4.90	7.43; 4.93; 4.36; 3.81; 3.31; 2.89; 2.55; 2.31; 2.15; 2.06; 6.35	7.02E−03; 4.88E−03; 6.00E−03; 7.02E−03; 7.65E−03; 7.70E−03; 7.15E−03; 6.21E−03; 5.12E−03; 4.11E−03; 3.22E−03
12	4	10	18.9	17.3	8.2	192.87	33.87; 24.83; 22.15; 19.66; 17.38; 15.32; 13.49; 11.89; 10.50; 9.30	9.37; 5.76; 5.12; 4.51; 3.96; 3.45; 3.01; 2.62; 2.30; 6.77	7.42E−03; 4.35E−03; 4.31E−03; 4.18E−03; 3.96E−03; 3.66E−03; 3.30E−03; 2.90E−03; 2.49E−03; 2.10E−03
13	4	10	16.5	16.6	13.5	221.76	30.59; 21.66; 19.07; 16.63; 14.41; 12.45; 10.75; 9.29; 8.03; 6.91	7.64; 4.43; 3.86; 3.33; 2.86; 2.46; 2.14; 1.91; 1.75; 6.67	6.26E−03; 3.19E−03; 3.43E−03; 3.56E−03; 3.54E−03; 3.35E−03; 3.02E−03; 2.63E−03; 2.21E−03; 1.83E−03
14	4	12	16.6	16.4	15.0	250.41	30.89; 22.89; 20.10; 17.52; 15.19; 13.11; 11.29; 9.73; 8.40; 7.28; 6.33; 5.54	8.76; 5.77; 5.06; 4.39; 3.78; 3.23; 2.74; 2.32; 1.97; 1.67; 1.43; 6.88	1.15E−02; 5.82E−03; 5.40E−03; 4.94E−03; 4.45E−03; 3.93E−03; 3.39E−03; 2.85E−03; 2.34E−03; 1.88E−03; 1.48E−03; 1.15E−03
15	1	11	14.9	14.6	–	272.00	28.69; 21.89; 19.80; 17.74; 15.75; 13.87; 12.13; 10.56; 9.17; 7.95; 6.90	6.77; 4.65; 4.22; 3.80; 3.38; 2.99; 2.64; 2.32; 2.05; 1.82; 6.68	7.23E−03; 3.32E−03; 3.45E−03; 3.61E−03; 3.75E−03; 3.83E−03; 3.83E−03; 3.71E−03; 3.48E−03; 3.16E−03; 2.79E−03

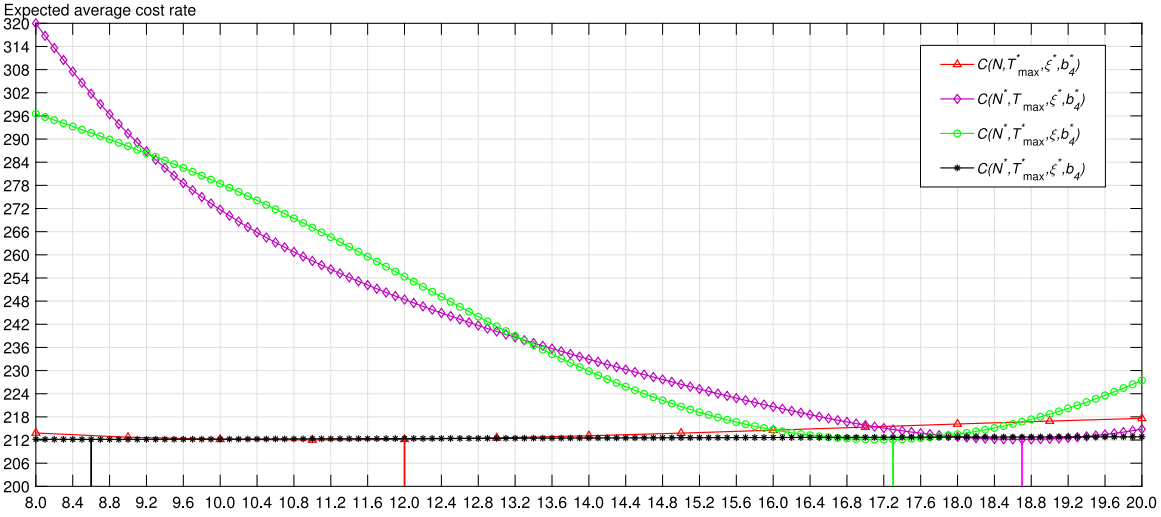


Fig. 7. Curves of expected average cost rate versus N, $T_{\max}$, ξ and b_4 (Machine 2 and inspection Policy 4).

Meanwhile, the computational results of different maintenance policies are shown in Table 3. More precisely, (1) by comparing with the CBM policy that proposed by Li et al. (2020) (the PM cost is C^0_{PM}), we can find that the proposed CBPM model with non-periodic concave decreasing inspection is better; and (2) by comparing with the CBM policy that ignored the impact of imperfect PM and only considered the optimal replacement (e.g., Dong et al., 2021; Huynh et al., 2011, 2012), we can observe that the maintenance policy without considering the imperfect PM (i.e., $N = 1$) is far inferior to the one with imperfect PM (i.e., the proposed maintenance model). To sum up, the proposed maintenance policy is better than some traditional maintenance policies.

6. Performance analysis of the proposed HMOEA algorithm

In this section, the testing instances and evaluation metric are first introduced. Second, the Taguchi method is employed to find the best combination of four key parameters. Third, the effect of different order sorting rules is analyzed. Fourth, the effectiveness of improved operators is verified by internal analysis of the algorithm. Finally, the superiority of the proposed algorithm is demonstrated by comparing with other intelligent algorithms.

6.1. Testing instances and evaluation metric

To test the overall performance of the proposed HMOEA algorithm, eight instances[1] (Ins) are employed (An, Chen, Gao, Zhang, et al., 2022), which can be further classified into two classes, i.e., small-scale problem (Instances 1–4) and large-scale problem (Instances 5–8). For each instance, the job inserting sequence (Seq), inserting time, inserting number, raw material cost, selling price, due date and deadline are separately displayed in Appendix H, and the maintenance planning for each machine comes from Table 2. Meanwhile, after each (re)scheduling, only one optimal scheduling solution is selected

[1] The dataset is available at: https://www.researchgate.net/publication/353938724_Rescheduling_dataset_new_job_insertion

Table 3
Optimal maintenance decisions for Li et al. (2020) maintenance policy and only replacement policy ($N = 1$).

M_k	Li's maintenance model					Only replacement model ($N = 1$)						M_k	Li's maintenance model					Only replacement model ($N = 1$)						M_k	Li's maintenance model					Only replacement model ($N = 1$)						M_k	Li's maintenance model					Only replacement model ($N = 1$)					
	N^*	$T^*_{\max}$	ξ^*	ACR*	Dev	N^*	$T^*_{\max}$	ξ^*	b^*_o	ACR*	Dev		N^*	$T^*_{\max}$	ξ^*	ACR*	Dev	N^*	$T^*_{\max}$	ξ^*	b^*_o	ACR*	Dev		N^*	$T^*_{\max}$	ξ^*	ACR*	Dev	N^*	$T^*_{\max}$	ξ^*	b^*_o	ACR*	Dev		N^*	$T^*_{\max}$	ξ^*	ACR*	Dev	N^*	$T^*_{\max}$	ξ^*	b^*_o	ACR*	Dev
1	2	11.8	19.4	288.95	68.14	1	18.0	18.5	4.4	284.24	63.43	6	3	19.0	18.8	287.34	68.17	1	21.2	20.8	4.4	304.32	85.15	11	2	11.5	19.8	328.35	80.93	1	18.0	19.0	4.3	336.24	88.81
2	3	13.3	20.7	289.39	77.22	1	20.1	19.8	4.3	303.10	90.93	7	1	17.4	14.2	343.10	93.41	1	15.4	17.4	4.7	334.11	84.42	12	2	18.4	17.6	260.74	67.87	1	20.1	19.2	4.4	263.81	70.94
3	2	18.1	17.1	315.83	82.79	1	19.6	18.9	4.5	322.25	89.21	8	2	17.1	17.6	276.48	47.67	1	18.7	19.5	4.3	284.87	56.05	13	2	16.9	16.9	293.57	71.81	1	18.3	18.6	4.5	300.08	78.33
4	2	16.9	15.0	323.37	75.69	1	16.0	18.0	4.5	325.56	77.88	9	1	10.6	18.9	269.93	64.14	1	18.3	17.8	4.7	260.62	54.82	14	2	17.5	16.4	325.53	75.13	1	18.4	18.3	4.5	333.05	82.65
5	2	11.6	19.6	293.32	68.43	1	18.0	18.7	4.4	297.85	72.97	10	2	12.1	19.0	359.36	101.52	1	18.7	18.0	4.5	354.52	96.68	15	2	10.8	18.4	372.62	94.68	1	16.9	17.6	4.6	368.48	90.53

Note: Dev = ACR* of other maintenance policy – ACR* of the proposed CBPM with non-periodic concave decreasing inspection.

according to the TOPSIS method (Tavana et al., 2016). To facilitate the evaluation of the optimal scheduling schemes for multiple runs, the average ($\overline{f_p}$) and standard deviation (stdf_p) of each objective in every scheduling stage are first calculated, and then they (i.e., $g_i \in [\overline{f_1}, \text{std}f_1, \overline{f_2}, \text{std}f_2, \overline{f_3}, \text{std}f_3]$) are normalized by the following formula:

$$g_i^n = (g_i^{\max} - g_i)/(g_i^{\max} - g_i^{\min}), \quad i = 1, 2, \ldots, 6, \tag{21}$$

where $g_i^{\max}$ and $g_i^{\min}$ are the maximum and minimum values of the ith indicator. If $g_i^{\max} = g_i^{\min}$, let $g_i^n = 1$. Based on this, we define fit$=\sum_{i=1}^{6} g_i^n$ as the fitness value (i.e., evaluation metric), and the larger is the better.

Additionally, all programs are coded and implemented in MATLAB R2017b and run on an Intel(R) Core(TM) i7-8700 CPU @ 3.20 GHz PC with 16 GB RAM memory. Each instance independently run twenty times, and then the means, standard deviations and fits can be calculated (in this section, at each rescheduling point, we randomly choose a rescheduling strategy from the rescheduling strategy pool).

6.2. Parameter setting

In the proposed algorithm, there are four input parameters, i.e., population size N, crossover probability P_c, mutation probability P_m and RV adjustment parameter f_r. Among them, we define each parameter has three different levels, i.e., $N \in \{21, 28, 36\}$, $P_c \in \{0.7, 0.8, 0.9\}$, $P_m \in \{0.05, 0.1, 0.15\}$ and $f_r \in \{0.1, 0.2, 0.3\}$. To obtain the best combination of four key parameters, the Taguchi method of design-of-experiments (Montgomery, 2005) is carried out based on Instances 3 and 6 (since they can separately represent the small-scale and large-scale problems). By running each parameter combination twenty times (the maximum number of iterations is $G = 10$), the means and standard deviations are reported in Appendix I. For easy evaluation, the sum of fitness values (i.e., $\sum$fit) is denoted as the response value to assess different parameter settings, and the analysis results are reported in Figs. 8 and 9, respectively. According to the maximum signal-to-noise ratios (SNRs) criterion, the best parameter combination for small-scale problem is $N = 21$, $P_c = 0.8$, $P_m = 0.1$ and $f_r = 0.1$, and the importance of four parameters is ranked as 4, 3, 2 and 1; the best parameter combination for large-scale problem is $N = 36$, $P_c = 0.8$, $P_m = 0.15$ and $f_r = 0.3$, and the importance of four parameters is ranked as 1, 2, 4 and 3. That is to say, the optimal parameter setting and the importance of these parameters are different for different scale problems.

6.3. Effect analysis of different order sorting rules

Since there is no existing optimal order sorting rule that can make the multi-order sorting optimal in a dynamic environment, we define fifteen rules (see Table 4) to find one or more appropriate order sorting rules for different scale problems (see lines 1 and 39 in Algorithm 1). Among them, Rules 1–3 are widely used ones in the literature; Rules 4–5 are the slack or tolerance that come from Guinet and Solomon (1996); Rules 6–7 are the average production cost rate (APCR) that is defined by the raw material cost, average processing cost and average tardy cost; and Rules 8–15 are the combinations of tolerance and APCR. In order to test the performance of these order sorting rules, Instances 3 and 6 are employed, and the evaluation results are reported in Fig. 10 ($\sum$fit as the response value and the computational results are reported in Appendix J). As can be seen from Fig. 10, the optimal sorting method

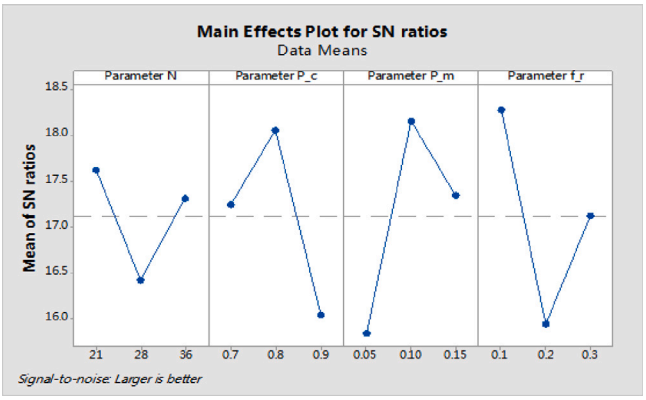


Fig. 8. Signal-to-noise ratios (SNR) for Instance 3.

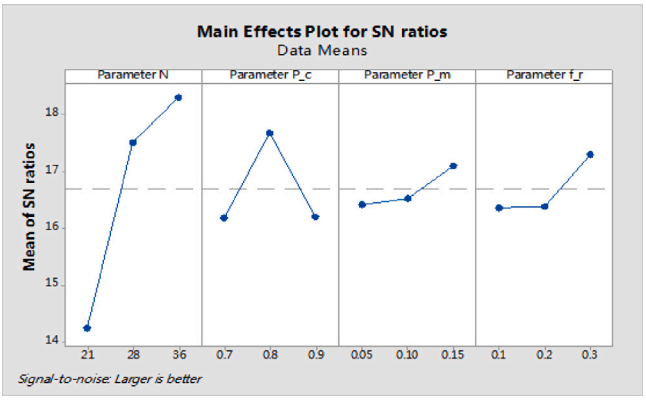


Fig. 9. Signal-to-noise ratios (SNR) for Instance 6.

is Rule 7 if a small-scale problem is solved; otherwise, Rule 14 is optimal. For the small-scale problem, the workload of the production machines is low, so we mainly consider the impact of production cost. For the large-scale problem, the workload is high, and reducing the tardiness penalty has become the main contradiction, so the sorting rule changes to the tolerance. Meanwhile, by comparing Rule 14 with Rules 5 and 6, we can find that the mix rule is significantly better than the separate ones.

6.4. Effect analysis of the improved operators

To verify the effectiveness of the improved operators, seven variants based on HMOEA are proposed, which are: (1) the initialization method comes from (Zhang et al., 2011), named HMOEA-GLR; (2) the initialization method comes from (An, Chen, Gao, Li, & Zhang, 2022), named HMOEA-GSC; (3) the N5 structure is removed from the hybrid LS, named HMOEA-N5; (4) the OL strategy is removed from the hybrid LS, called HMOEA-OL; (5) the processing time window is removed from the hybrid LS, called HMOEA-Time window; (6) the tournament selection operator is used to select individuals for crossover, called HMOEA-Tournament section; and (7) the random selection operator is employed

Table 4
Fifteen order sorting rules.

Rule 1: Earliest deadline first (EDF)	Rule 2: Shortest average processing time (APT) first	Rule 3: Longest APT first
Rule 4: Maximum-slack-time first	Rule 5: Minimum-slack-time first	Rule 6: Maximum APCR first
Rule 7: Minimum APCR first	Rule 8: Tolerance descend+ APCR descend	Rule 9: Tolerance descend+ APCR ascend
Rule 10: Tolerance ascend+ APCR descend	Rule 11: Tolerance ascend+ APCR ascend	Rule 12: APCR descend+Tolerance descend
Rule 13: APCR ascend+Tolerance descend	Rule 14: APCR descend+Tolerance ascend	Rule 15: APCR ascend+Tolerance ascend

Note: Slack or Tolerance$=\overline{D_i}-\sum_{j=1}^{n_i}\sum_{k=1}^{|\mathrm{AMS}_{ij}|}\frac{P_{ijk}}{|\mathrm{AMS}_{ij}|}-R_i$; APCR$=\left(\mathrm{RMC}_i+\sum_{j=1}^{n_i}\sum_{k=1}^{|\mathrm{AMS}_{ij}|}\frac{P_{ijk}\cdot \mathrm{PC}_k}{|\mathrm{AMS}_{ij}|}+\varpi_i\right)/\sum_{j=1}^{n_i}\sum_{k=1}^{|\mathrm{AMS}_{ij}|}\frac{P_{ijk}}{|\mathrm{AMS}_{ij}|}$.

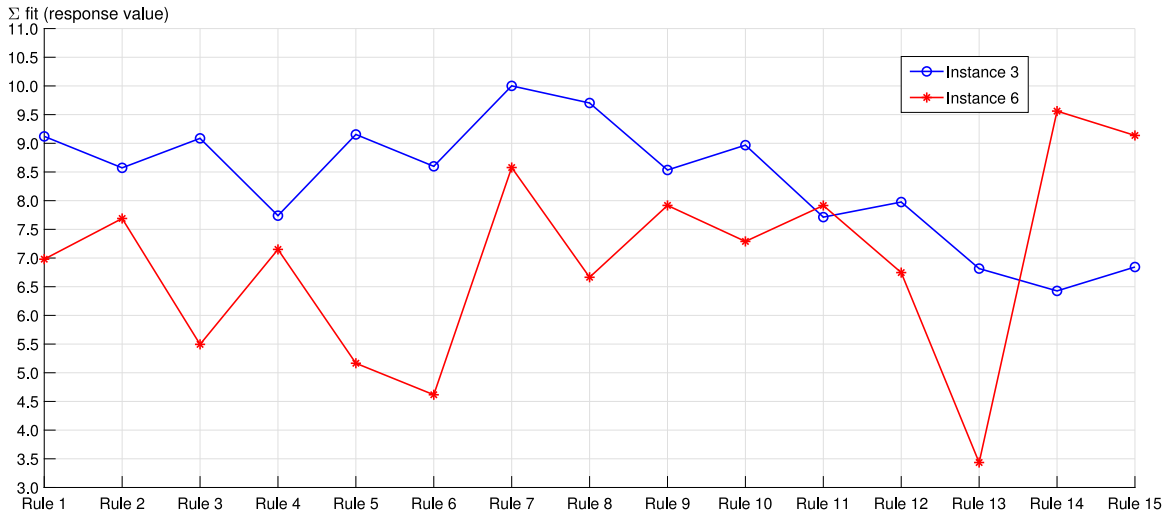


Fig. 10. The performance of different order sorting rules for Instances 3 and 6.

Table 5
Computational results of the improved operators and other intelligent algorithms.

Algorithm	Instance 3		Instance 6		Algorithm	Instance 3		Instance 6	
	∑fit	Gap	∑fit	Gap		∑fit	Gap	∑fit	Gap
HMOEA-GLR	9.68	−10.99	13.43	−19.83	NSGA-II	6.00	−44.84	14.30	−14.64
HMOEA-GSC	9.57	−12.00	16.32	−2.64	NSGA-III	7.81	−28.17	11.38	−32.09
HMOEA-N5	7.13	−34.44	16.52	−1.40	ANSGA-III-ARV	9.30	−14.43	15.90	−5.13
HMOEA-OL	6.48	−40.37	16.46	−1.76	INSGA-III-ARV	8.70	−20.00	15.77	−5.87
HMOEA-Time window	10.15	−6.66	16.41	−2.10	HMOEA	10.87	–	16.76	–
HMOEA-Tournament section	9.83	−9.58	16.79	0.18					
HMOEA-Random section	11.12	2.27	16.75	−0.02					

Note: Gap = (∑fit from other algorithms −∑fit from HMOEA)×100/∑fit from HMOEA.

to select individuals for crossover, denoted HMOEA-Random section. By testing above algorithms on Instances 3 and 6, the running results are reported in Appendix K, and the sum of fits and their gaps are displayed in Table 5. From Table 5, we can find that (1) the proposed hybrid initialization method is significantly better than other two conventional methods since their gaps are all negative; (2) each constituent element in the hybrid LS plays an important role, since removing any element will cause its overall performance to be worse, and this phenomenon is more pronounced on small-scale problem; and (3) there is no significant difference among three selection operators, which may be due to a large number of iterations and local searches that make up for the difference in selection operators. On the whole, these improved operators play a positive and significant role in improving the overall performance of the proposed algorithm.

6.5. Comparison with other intelligent algorithms

To provide a comprehensive and reliable comparison, the proposed HMOEA algorithm is compared with four state-of-the-art intelligent algorithms, which are: NSGA-II (Deb et al., 2002), NSGA-III (Deb & Jain, 2014), NSGA-III/ARV (An, Chen, Gao, Li, & Zhang, 2022) and INSGA-III-ARV (An, Chen, Gao, Zhang, et al., 2022). For a fair comparison, the crossover, mutation and parameter setting of these four algorithms remain the same as HMOEA. Similarly, Instances 3 and 6 are also employed to test the overall performance. The tested means and standard deviations of five algorithms are displayed in Appendix K, and the sum of fits and their gaps are reported in Table 5. According to the calculation results, there are several conclusions that can be obtained: (1) NSGA-II and NSGA-III are significantly worsen than HMOEA since the gaps are the most negative, which further demonstrates the values of both the designed hybrid LS and RV update mechanism; (2) HMOEA outperforms NSGA-III/ARV and INSGA-III/ARV on small-scale problem, while the gap is smaller on large-scale problem; and (3) HMOEA has a better overall performance among these cutting-edge algorithms, since their gaps are all negative.

7. Statistical analysis of rescheduling strategies under different scenarios

In this section, three rescheduling factors and twenty-seven rescheduling scenarios are first constructed. Second, three response

Table 6
The level setting and division of different rescheduling factors.

Level	Rescheduling strategy	Remaining task size (RTS)	Inserted task size (ITS)
1	S1	S ($\lfloor 0.05N_0 \rfloor \le$ RTS $\le \lfloor 0.3N_0 \rfloor$)	S ($\lfloor 0.05N_0' \rfloor \le$ ITS $\le \lfloor 0.3N_0' \rfloor$)
2	S2	M ($\lfloor 0.35N_0 \rfloor \le$ RTS $\le \lfloor 0.6N_0 \rfloor$)	M ($\lfloor 0.35N_0' \rfloor \le$ ITS $\le \lfloor 0.6N_0' \rfloor$)
3	S3	B ($\lfloor 0.65N_0 \rfloor \le$ RTS $\le \lfloor 0.9N_0 \rfloor$)	B ($\lfloor 0.65N_0' \rfloor \le$ ITS $\le \lfloor 0.9N_0' \rfloor$)

indicators are designed to evaluate the rescheduling performance under different rescheduling scenarios. Third, a comprehensive statistical analysis is performed to analyze the performance of different rescheduling strategies. Finally, the optimal rescheduling strategies under different scenarios are recommended.

7.1. Rescheduling factors and rescheduling scenarios construction

In this work, three rescheduling factors are considered: (1) rescheduling methods, including S1, S2 and S3; (2) remaining task size (RTS, i.e., N_0'), including the level of small (S), middle (M) and big (B), which is determined by the ratio of RTS to previous total scheduling operations (N_0); and (3) inserted task size (ITS), including the level of small (S), middle (M) and big (B), which is determined by the ratio of ITS to total remaining unstated operations (N_0'). Based on these, the level setting and division of three factors are shown in Table 6. For simplicity, without considering the rescheduling strategy, nine rescheduling scenarios can be expressed in the form of two variables (RTS,ITS), e.g., the rescheduling scenario (S,S) indicates the remaining task size and newly inserted task size are all small.

Since there are three levels in each rescheduling factor, so they can form $3^3 = 27$ scenarios. In order to cover all scenarios, on the basis of Instances 3 and 6 in Appendix H, we duplicate Jobs 11–13 and Jobs 16–21 as a candidate jobset (i.e., Jobs 11–16 and Jobs 16–27) of the inserted orders. Therefore, two new instances are constructed, i.e., 16×6 (renamed as Instance $3'$) and 27×10 (renamed as Instance $6'$). To highlight the statistical significance, each scenario will be tested 20 times with one rescheduling. Thus, the whole experiment consists $2 \times 27 \times 20 = 1080$ experimental units for the two test instances.

7.2. Response indicators

According to the optimization objectives, three response indicators are designed to measure the completion time (f_1), instability (f_2) and total production cost (f_3) in the context of dynamic order arrivals, which are described as follows:

(1) Ratio of completion time (RCT)

$$\mathrm{RCT}(S_0, S_N) = f_1(S_N)/f_1(S_0). \tag{22}$$

where S_0 is the previous scheduling scheme and S_N is the rescheduling scheme. The smaller the RCT is, the smaller the fluctuation of makespan is.

(2) Instability (INS)

$$\mathrm{INS}(S_N) = f_2(S_N). \tag{23}$$

where S_N is a constructed rescheduling plan. The smaller the INS is, the better the stability is.

(3) Ratio of total production cost (RTPC)

$$\mathrm{RTPC}(S_0, S_N) = f_3(S_N)/f_3(S_0). \tag{24}$$

where S_0 is the previous scheduling plan and S_N is the constructed rescheduling plan. The smaller the RTPC is, the smaller the augment in production cost is.

7.3. Statistical analysis of the computational results on different indicators

To identify the performance of different rescheduling strategies in different rescheduling scenarios, we analyze it from the following three aspects:

(1) Statistical analysis about RCT

By running the above two newly constructed instances, the computational results about RCT are shown in Figs. 11 and 12, respectively. Through analyzing the computational results, we can find that (1) when the scale of the remaining task is almost the same [e.g., scenario (S,*)], as the number of inserted tasks increases, the completion time is gradually increased, i.e., (S,S)<(S,M)<(S,B); and (2) when the ratio of the inserted task size to the remaining task size is almost the same [e.g., scenario (*,S)], as the amount of remaining tasks increases, the number of operations involved in rescheduling increases, so the completion time tends to increase in most rescheduling strategies [e.g., in S3, there exists (S,S)<(M,S)<(B,S)]. Meanwhile, through Friedman test on RCT (the significance level is 0.1), it is found that there are significant differences among three rescheduling strategies since their p values are all less than 0.1 (p = 1.3749E-20 on Instance $3'$, and p = 9.7708E-18 on Instance $6'$). Based on these, we further test the differences between S1 and S3 on Instance $3'$ and between S1 and S2 on Instance $6'$ as they have similar properties, and the test results show that they also have significant differences. Finally, by combining Figs. 11 and 12 with the statistical analysis results, we can conclude that S2 is significantly better than other two rescheduling strategies on RCT.

(2) Statistical analysis about INS

Similar to RCT, the statistical results on INS are displayed in Figs. 13 and 14. According to the running results, except for S1, the instability of other two rescheduling methods is basically linearly related to the remaining task size or the total number of rescheduling operations. For example, when the ratios of the inserted task size to the remaining task size are in the same level, the instability gradually increases as the amount of remaining tasks increasing, e.g., (S,S)<(M,S)<(B,S). At the same time, the instability is not significantly related to the inserted task size when the remaining task size is almost the same, e.g., (S,S)≈(S,M)≈(S,B). Furthermore, nonparametric Friedman test with 90% confidence level is carried out among three rescheduling methods, and the test p-values are 1.3549E−37 and 8.1940E−40 on two instances (i.e., there are significant differences among three rescheduling strategies). Additionally, S2 fluctuates more than S3, and their fluctuation increases with the expansion of the scale of the test problem. To sum up, S1 significantly outperforms other two rescheduling strategies on INS.

(3) Statistical analysis about RTPC

Analogous to above two indicators, the computational results on RTPC are plotted in Figs. 15 and 16. According to the calculation results, we can observe that (1) insertion of new jobs will lead to increased total production cost due to RTPC > 1, and when the ratio of the inserted task scale to the remaining task scale is almost the same, the earlier the rescheduling occurs, the lower the production cost, e.g., (S,S)>(M,S)>(B,S); (2) when the remaining task size is almost the same, the more tasks inserted, the better the optimization effect on the production cost [e.g., (S,S)>(S,M)>(S,B)], which may be the result of sacrificing other two objectives; (3) S1 is significantly inferior to S2 and S3, since RTPC is relatively large and the p-values of the Friedman test on RTPC are all less than 0.1 (p = 2.3784E−30 on Instance $3'$,

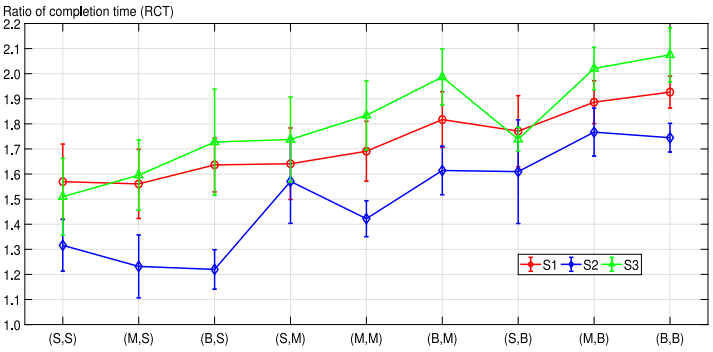


Fig. 11. Ratio of completion time for Instance 3′.

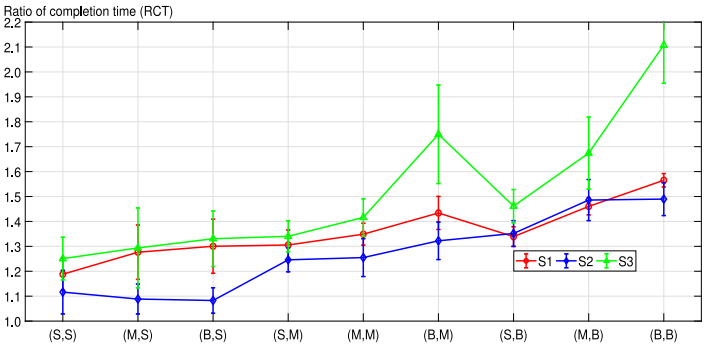


Fig. 12. Ratio of completion time for Instance 6′.

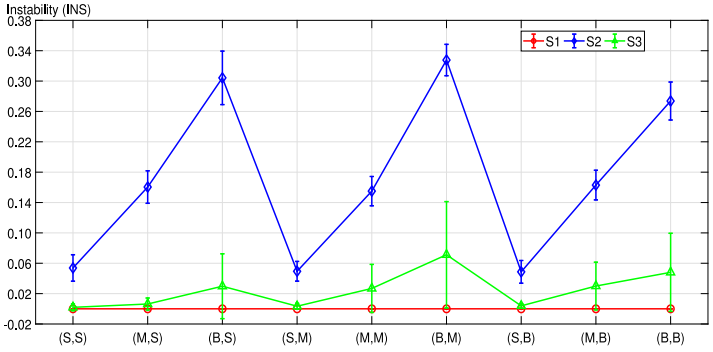


Fig. 13. Instability for Instance 3′.

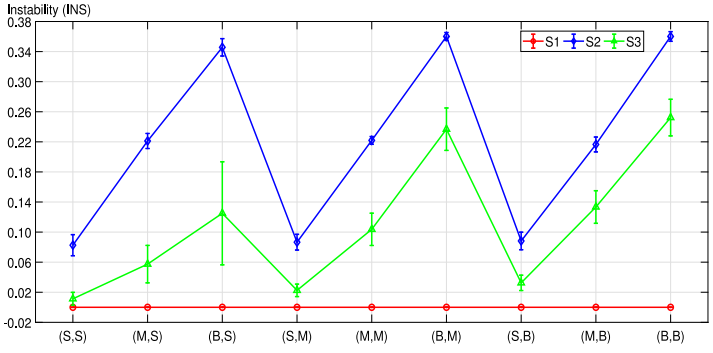


Fig. 14. Instability for Instance 6′.

and p = 5.4869E−31 on Instance 6′); and (4) there is no significant difference between S2 and S3, since their p values of nonparametric Kruskal–Wallis test with 90% confidence level are all greater than 0.1 (p = 0.2596 on Instance 3′, and p = 0.2454 on Instance 6′). To sum up, S2 and S3 significantly outperform S1 on RTPC.

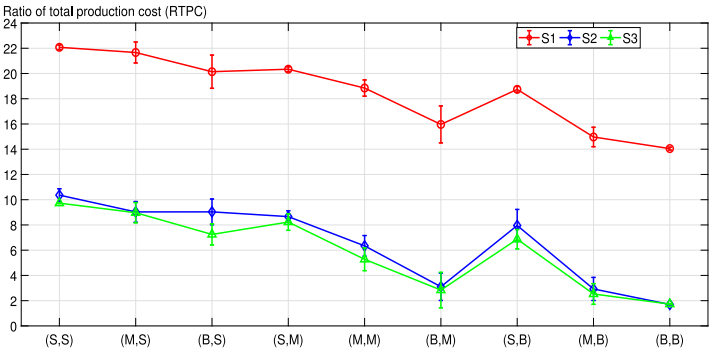


Fig. 15. Ratio of total production cost for Instance 3′.

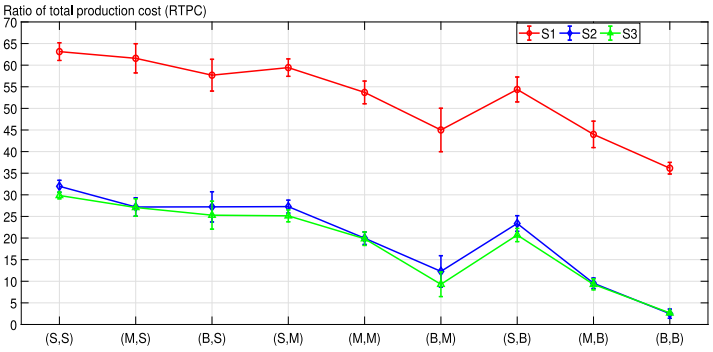


Fig. 16. Ratio of total production cost for Instance 6′.

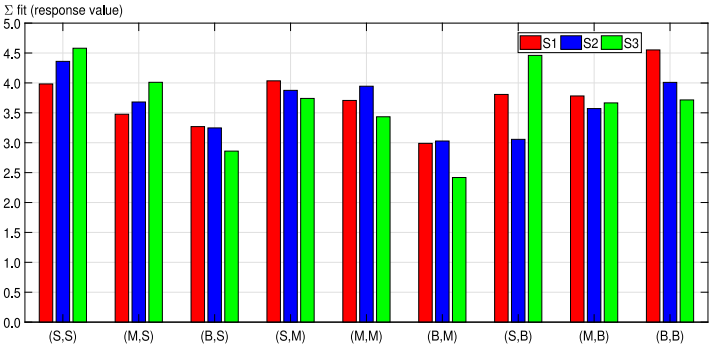


Fig. 17. Response value for Instance 3′.

7.4. Discussion on the optimal rescheduling strategies under different scenarios

Through the analysis in the previous subsection, it is found that different rescheduling strategies have significant differences in different indicators. For example, S2 is dominant in RCT indicator, S1 is dominant in INS indicator, and S2 and S3 are dominant in RTPC indicator. In order to comprehensively evaluate the differences in the overall performance of different rescheduling strategies, we utilize the sum of fit (i.e., $\sum$fit) as response value, and the statistical results on different instances are reported in Figs. 17 and 18, respectively. From Figs. 17 and 18, it can be observed that the response value fluctuates more on Instance 6′ than on Instance 3′. According to the principle of maximizing the response value, the optimal rescheduling strategies (ORSs) in different rescheduling scenarios are recommended in Table 7. As can be seen from Table 7, S1 plays a dominant role in small-scale problem (i.e., Instance 3′), but in large-scale problem (i.e., Instance 6′), S2 and S3 play dominant roles.

8. Sensitivity analysis on the adaptive rescheduling problem

Based on the test results of Sections 5–7, the influence of three parameters [including initial order acceptance ratios (γ_0 and γ_c), rejection penalty (λ) and instability threshold (θ)] on the adaptive rescheduling problem is further analyzed.

8.1. Sensitivity analysis on the initial addition ratios

In order to analyze the impact of different initial order acceptance ratios on both the average number (AN) of accepted orders and overall performance (i.e., fit), we vary each initial addition ratio between 0.2 and 0.8 with step size 0.2 (other two parameters are defined as: $\lambda = 10$ and $\theta = 1$). Eight instances constructed in Section 6.1 are employed, and the calculated means and standard deviations of different initial addition ratios are reported in Appendix L. Based on the mean and standard deviation, the fitness value (i.e., fit) and corresponding gap in each scheduling stage can be obtained, as shown in Table 8. From Table 8, we can find that the average fits are almost the same under

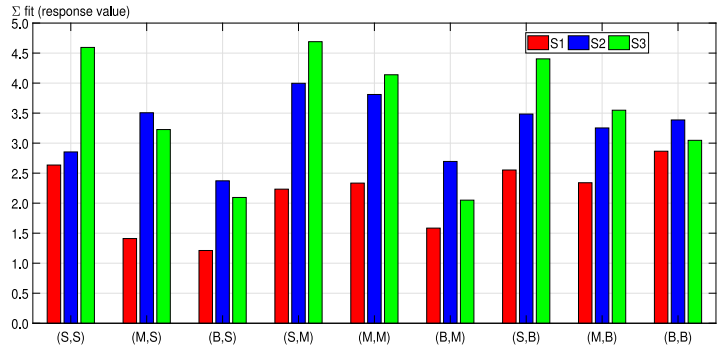


Fig. 18. Response value for Instance 6′.

Table 7
Optimal rescheduling strategies (ORSs) for different rescheduling scenarios.

Instance 3′						Instance 6′					
Scenario	ORS	Scenario	ORS	Scenario	ORS	Scenario	ORS	Scenario	ORS	Scenario	ORS
(S,S)	S3	(S,M)	S1	(S,B)	S3	(S,S)	S3	(S,M)	S3	(S,B)	S3
(M,S)	S3	(M,M)	S2	(M,B)	S1	(M,S)	S2	(M,M)	S3	(M,B)	S3
(B,S)	S1	(B,M)	S2	(B,B)	S1	(B,S)	S2	(B,M)	S2	(B,B)	S2

Table 8
Computational results of eight test instances under different initial addition ratios.

Ins	Seq	$\gamma_0 = 0.2$		$\gamma_0 = 0.4$			$\gamma_0 = 0.6$			$\gamma_0 = 0.8$			$\gamma_c = 0.2$		$\gamma_c = 0.4$			$\gamma_c = 0.6$			$\gamma_c = 0.8$		
		AN	fit	AN	fit	Gap	AN	fit	Gap	AN	fit	Gap	AN	fit	AN	fit	Gap	AN	fit	Gap	AN	fit	Gap
1	0	4.25	6.00	4.75	5.98	−0.40	5	6.00	0	5	5.97	−0.52	5	6.00	5	6.00	0	5	6.00	0	5	6.00	0
	1	0	5.98	0	5.96	−0.40	0	5.98	0	0	5.95	−0.52	0	5.98	0	5.98	0	0	5.98	0	0	5.98	0
	2	0	5.96	0	5.94	−0.40	0	5.96	0	0	5.93	−0.52	0	5.96	0	5.96	0	0	5.96	0	0	5.96	0
2	0	7.25	5.75	7.75	5.67	−1.46	8	5.71	−0.79	8	5.71	−0.65	7.5	5.73	7.75	5.70	−0.53	7.75	5.62	−1.77	7.5	5.65	−1.28
	1	0	5.70	0	5.61	−1.47	0	5.65	−0.80	0	5.66	−0.65	0	5.67	0	5.64	−0.53	0	5.57	−1.78	0	5.60	−1.29
	2	0	5.64	0	5.56	−1.49	0	5.60	−0.80	0	5.60	−0.66	0	5.62	0	5.59	−0.54	0	5.51	−1.80	0	5.54	−1.31
3	0	9.25	5.34	9.75	5.42	1.59	10	5.52	3.40	10	5.53	3.74	9.5	5.43	10	5.55	2.16	10	5.60	3.23	10	5.55	2.21
	1	0	5.29	0	5.37	1.60	0	5.47	3.43	0	5.49	3.77	0	5.38	0	5.50	2.18	0	5.55	3.26	0	5.50	2.23
	2	0	5.23	0	5.31	1.62	0	5.41	3.47	0	5.42	3.82	0	5.32	0	5.44	2.21	0	5.49	3.29	0	5.44	2.25
	3	0	5.16	0	5.25	1.64	0	5.34	3.51	0	5.36	3.86	0	5.26	0	5.37	2.23	0	5.43	3.33	0	5.38	2.28
4	0	10	5.67	10	5.64	−0.54	10	5.61	−1.19	10	5.68	0.08	10	5.67	10	5.68	0.24	10	5.71	0.71	10	5.52	−2.58
	1	0	5.57	0	5.54	−0.55	0	5.50	−1.22	0	5.58	0.08	0	5.57	0	5.58	0.24	0	5.61	0.72	0	5.42	−2.63
	2	0	5.45	0	5.42	−0.56	0	5.38	−1.24	0	5.46	0.08	0	5.45	0	5.46	0.25	0	5.49	0.74	0	5.30	−2.68
5	0	15	5.15	15	5.18	0.68	15	5.09	−1.08	15	5.09	−1.07	15	5.11	15	5.10	−0.20	15	5.19	1.56	15	5.23	2.28
	1	0	5.01	0	5.05	0.70	0	4.96	−1.11	0	4.96	−1.10	0	4.98	0	4.97	−0.20	0	5.06	1.60	0	5.10	2.34
	2	0	4.56	0	4.87	6.75	0	4.78	4.77	0	4.78	4.77	0	4.80	0	4.79	−0.21	0	4.88	1.66	0	4.92	2.43
6	0	15	5.25	15	5.21	−0.72	15	5.19	−1.25	15	5.31	1.08	15	5.20	15	5.25	0.90	15	5.31	2.10	15	5.24	0.75
	1	0.5	3.52	0	4.97	41.06	0.5	3.24	−7.93	0	5.06	43.74	0.5	3.26	0	5.01	53.47	0	5.07	55.40	0	5.00	53.23
	2	0.5	2.48	0	4.63	87.05	0.5	2.69	8.45	0	4.73	90.87	0.5	2.70	0	4.67	72.63	0	4.73	74.94	0	4.66	72.33
	3	0.25	2.12	0	4.27	101.75	0.25	2.05	−2.92	0	4.36	106.22	0.25	2.07	0	4.31	107.66	0	4.37	110.68	0	4.30	107.27
7	0	20	4.79	20	4.74	−1.03	20	4.85	1.28	20	4.78	−0.32	20	4.88	20	4.86	−0.53	20	4.84	−0.79	20	4.65	−4.84
	1	0	4.54	0	4.49	−1.08	0	4.60	1.35	0	4.53	−0.34	0	4.63	0	4.61	−0.56	0	4.60	−0.83	0	4.40	−5.09
	2	0	4.28	0	4.23	−1.15	0	4.34	1.43	0	4.26	−0.36	0	4.37	0	4.34	−0.59	0	4.33	−0.88	0	4.13	−5.40
8	0	20	4.95	20	4.97	0.42	20	4.89	−1.10	20	4.97	0.55	20	4.92	20	4.98	1.29	20	4.96	0.79	20	4.85	−1.31
	1	0	4.72	0.25	3.31	−29.83	1	2.28	−51.74	0.25	3.08	−34.70	1	2.31	0	4.76	106.36	0	4.73	105.28	0	4.63	100.80
	2	0	4.45	0.5	2.83	−36.41	0.75	1.69	−62.02	0.5	2.52	−43.33	0.75	1.72	0	4.48	160.87	0	4.46	159.42	0	4.35	153.41
	3	0	4.27	0.5	2.36	−44.71	0.75	1.32	−68.96	0.5	2.06	−51.61	0.75	1.35	0	4.30	217.65	0	4.28	215.81	0	4.17	208.19
Average		3.78	4.92	3.83	4.95	4.54	3.95	4.63	−6.41	3.86	4.96	4.68	3.92	4.64	3.81	5.18	26.91	3.81	5.20	27.28	3.80	5.13	25.32

Note: Gap=[(fit from other addition rate γ_0)−(fit from addition rate $\gamma_0 = 0.2$)]×100/(fit from addition rate $\gamma_0 = 0.2$);
Gap=[(fit from other addition rate γ_c)−(fit from addition rate $\gamma_c = 0.2$)]×100/(fit from addition rate $\gamma_c = 0.2$).

different γ_0 or γ_c. In order to test the obtained results (i.e., fit) with different initial addition ratios are significantly different or not, the Friedman test results (the significance level is 0.1) are shown in Figs. 19 and 20, and the p-values under different γ_0 and γ_c are all 0.8633. That is to say, there is no significant difference under different addition ratios because the proposed algorithm can adaptively pursue the performance maximization.

8.2. Sensitivity analysis on the rejection penalty

To demonstrate the benefits of ROA, we compare it with all order acceptance (AOA) strategy under different rejection penalties (i.e., λ=5, 10, 15, 20 and 25). The calculated means and standard deviations of eight instances are displayed in Appendix M (other three parameters are defined as: $\gamma_0 = 0.6$, $\gamma_c = 0.2$ and $\theta = 1$), and then the fitness value (i.e., fit) and corresponding gap in each scheduling stage can be calculated, as shown in Table 9. The Friedman test with 90% confidence level is carried out between the fit values of AOA and the ones by ROA under any of the same penalties, and the test p-values are 6.6973E−04, 0.0833, 0.0011, 0.0343 and 5.3121E−05, respectively. According to the test results, there are three conclusions that can be obtained: (1) there is a significant difference between two order acceptance strategies under any of the same penalties, and accepting all orders may result in reduced overall performance since the average fit of ROA is greater than that of AOA (see the red line of Fig. 21); and (2) the fit value of AOA fluctuates more, but ROA is relatively stable, since the box length of the fit value of AOA is greater than that of the fit value of ROA in Fig. 21.

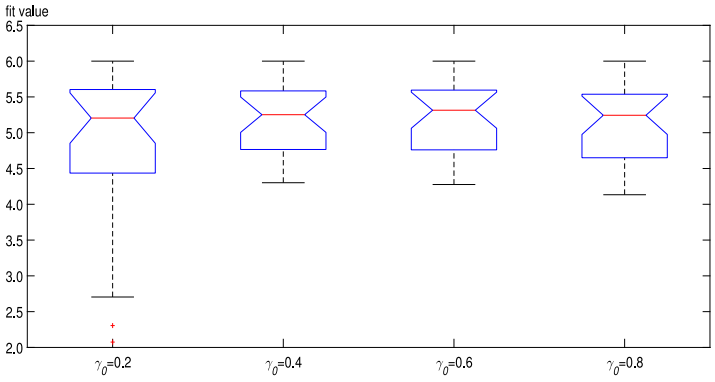


Fig. 19. Box-plot for different initial addition rate γ_0 (γ_c=0.2).

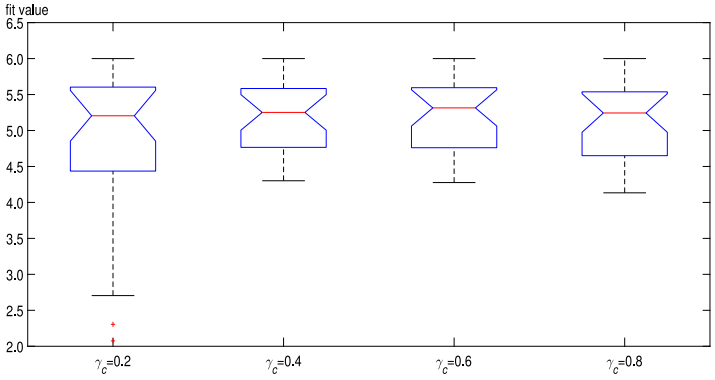


Fig. 20. Box-plot for different initial addition rate γ_c (γ_0=0.6).

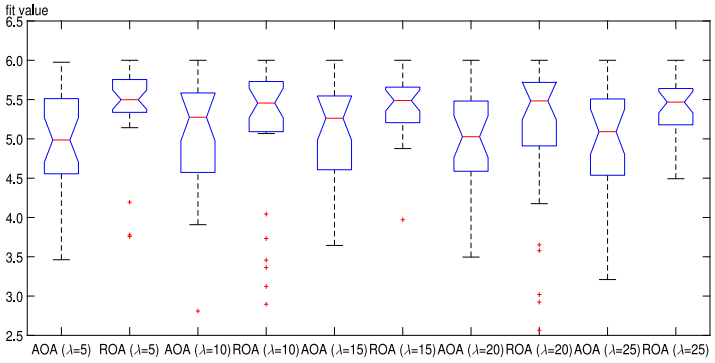


Fig. 21. Box-plot for different rejection penalty λ.

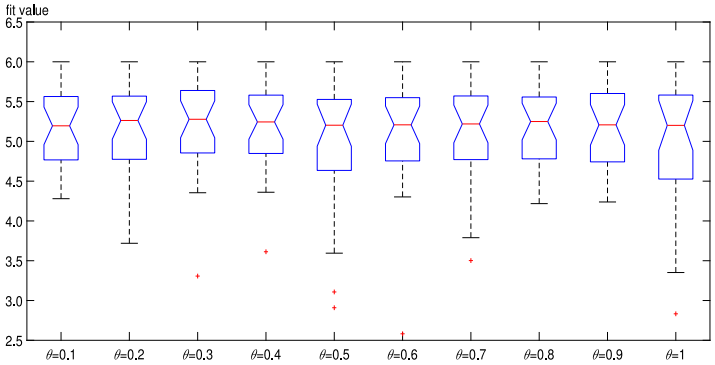


Fig. 22. Box-plot for different instability threshold θ.

Table 9
Computational results of eight test instances under different rejection penalties.

Ins	Seq	AOA ($\lambda = 5$)		ROA ($\lambda = 5$)			AOA ($\lambda = 10$)		ROA ($\lambda = 10$)			AOA ($\lambda = 15$)		ROA ($\lambda = 15$)			AOA ($\lambda = 20$)		ROA ($\lambda = 20$)			AOA ($\lambda = 25$)		ROA ($\lambda = 25$)		
		AN	fit	AN	fit	Gap	AN	fit	AN	fit	Gap	AN	fit	AN	fit	Gap	AN	fit	AN	fit	Gap	AN	fit	AN	fit	Gap
1	0	5	5.98	5	6.00	0.41	5	6.00	5	6.00	0	5	6.00	5	6.00	0	5	6.00	5	6.00	0	5	6.00	5	6.00	0
	1	1	5.49	0	6.00	9.20	1	5.80	0	5.99	3.28	1	5.65	0	5.99	5.93	1	5.74	0	5.99	4.28	1	5.64	0	5.98	6.13
	2	1	5.80	0	5.99	3.27	1	5.93	0	5.98	0.91	1	5.93	0	5.98	0.78	1	5.91	0	5.97	0.93	1	5.91	0	5.96	0.79
2	0	8	5.87	7.5	5.79	−1.35	8	5.88	8	5.81	−1.22	8	5.83	7.75	5.81	−0.32	8	5.86	7.75	5.78	−1.27	8	5.85	7.75	5.75	−1.71
	1	1	5.54	0	5.78	4.20	1	5.65	0	5.79	2.49	1	5.57	0	5.77	3.58	1	5.49	0	5.74	4.50	1	5.62	0	5.70	1.42
	2	1	4.89	0	5.77	17.90	1	4.96	0	5.76	16.23	1	5.42	0	5.74	5.94	1	5.42	0	5.69	4.99	1	5.26	0	5.64	7.13
3	0	10	5.67	10	5.57	−1.77	10	5.73	10	5.69	−0.62	10	5.68	9.75	5.61	−1.24	10	5.73	10	5.73	0.06	10	5.58	10	5.64	1.15
	1	1	5.51	0	5.56	0.76	1	5.52	0	5.67	2.74	1	5.38	0	5.58	3.77	1	5.42	0	5.69	4.97	1	5.23	0	5.59	6.99
	2	1	4.99	0	5.54	11.16	1	5.29	0	5.65	6.70	1	5.27	0	5.54	5.26	1	5.05	0	5.64	11.56	1	5.02	0	5.53	10.23
	3	1	4.98	0	5.53	10.98	1	5.29	0	5.62	6.28	1	5.26	0	5.50	4.64	1	5.00	0	5.59	11.70	1	4.97	0	5.47	10.00
4	0	10	5.78	10	5.77	−0.30	10	5.75	10	5.75	0.01	10	5.73	10	5.69	−0.69	10	5.70	10	5.79	1.70	10	5.78	10	5.71	−1.07
	1	2	5.15	0	5.74	11.51	2	5.35	0	5.71	6.70	2	5.30	0	5.63	6.07	2	5.26	0	5.71	8.58	2	5.05	0	5.61	11.10
	2	2	4.98	0	5.72	14.85	2	5.09	0	5.66	11.12	2	5.25	0	5.55	5.68	2	4.96	0	5.61	13.12	2	5.14	0	5.48	6.79
5	0	15	5.36	15	5.43	1.28	15	5.36	15	5.43	1.18	15	5.39	15	5.47	1.50	15	5.27	15	5.44	3.21	15	5.42	15	5.52	1.82
	1	2	4.51	0	5.40	19.75	2	4.57	0	5.37	17.45	2	3.64	0	5.39	47.89	2	3.71	0	5.33	43.54	2	4.37	0	5.38	23.16
	2	2	4.51	0	5.36	18.85	2	4.57	0	5.30	15.80	2	4.77	0	5.28	10.62	2	4.66	0	5.18	11.24	2	4.68	0	5.19	10.84
6	0	15	5.51	15	5.52	0.23	15	5.52	15	5.49	−0.60	15	5.52	15	5.52	0.03	15	5.47	15	5.52	0.95	15	5.44	15	5.47	0.62
	1	2	4.60	0	5.47	18.89	2	4.52	0.5	4.04	−10.62	2	4.56	0	5.37	17.62	2	4.59	0.25	4.18	−9.09	2	4.58	0	5.21	13.86
	2	2	4.60	0	5.40	17.43	2	4.71	0.5	3.73	−20.81	2	4.65	0	5.16	10.86	2	4.58	0.25	3.65	−20.32	2	4.49	0	4.87	8.27
	3	2	4.65	0	5.33	14.50	2	4.71	0.25	3.36	−28.61	2	4.65	0.5	3.97	−14.66	2	4.64	0.25	3.02	−34.97	2	4.77	0	4.49	−5.87
7	0	20	5.14	20	5.25	2.02	20	5.26	20	5.28	0.40	20	5.21	20	5.20	−0.22	20	5.23	20	5.23	−0.10	20	5.23	20	5.23	0.01
	1	3	3.86	0	5.20	34.66	3	3.91	0	5.18	32.48	3	3.94	0	5.04	27.97	3	3.87	0	5.02	29.77	3	4.09	0	4.97	21.50
	2	3	3.94	0	5.14	30.36	3	3.99	0	5.07	27.09	3	4.08	0	4.88	19.54	3	4.10	0	4.80	17.24	3	3.96	0	4.70	18.66
8	0	20	5.30	20	5.35	0.87	20	5.34	20	5.31	−0.67	20	5.27	20	5.36	1.64	20	4.94	20	5.40	9.32	20	5.17	20	5.40	4.31
	1	3	4.32	0.25	4.19	−2.81	3	4.23	1	3.46	−18.27	3	4.31	0	5.22	21.01	3	4.30	0.75	3.58	−16.72	3	4.09	0	5.17	26.42
	2	3	3.46	1	3.76	8.57	3	2.81	0.75	3.12	11.13	3	4.34	0	5.05	16.29	3	3.50	1.5	2.92	−16.40	3	3.21	0	4.88	52.00
	3	2	4.23	1.25	3.78	−10.70	2	4.26	0.75	2.90	−32.09	2	4.21	0	4.93	17.29	2	4.15	1.5	2.56	−38.21	2	4.21	0	4.69	11.34
Average		5.11	4.99	3.89	5.38	8.69	5.11	5.04	3.95	5.12	1.80	5.11	5.07	3.81	5.42	8.03	5.11	4.98	3.97	5.07	1.65	5.11	4.99	3.81	5.38	9.11

Note: Gap=[(fit from ROA)−(fit from AOA)]×100/(fit from AOA).

Table 10
Computational results of eight test instances under different instability thresholds.

Ins	Seq	$\theta = 0.1$		$\theta = 0.2$			$\theta = 0.3$			$\theta = 0.4$			$\theta = 0.5$			$\theta = 0.6$			$\theta = 0.7$			$\theta = 0.8$			$\theta = 0.9$			$\theta = 1$		
		AN	fit	AN	fit	Gap	AN	fit	Gap	AN	fit	Gap	AN	fit	Gap	AN	fit	Gap	AN	fit	Gap	AN	fit	Gap	AN	fit	Gap	AN	fit	Gap
1	0	5	6.00	5	6.00	0	5	6.00	0	5	6.00	0	5	6.00	0	5	6.00	0	5	6.00	0	5	6.00	0	5	6.00	0	5	6.00	0
	1	0	5.98	0	5.98	0	0	5.98	0	0	5.98	0	0	5.98	0	0	5.98	0	0	5.98	0	0	5.98	0	0	5.98	0	0	5.98	0
	2	0	5.96	0	5.96	0	0	5.96	0	0	5.96	0	0	5.96	0	0	5.96	0	0	5.96	0	0	5.96	0	0	5.96	0	0	5.96	0
2	0	7.75	5.70	7.75	5.65	−0.76	8	5.79	1.73	7.25	5.73	0.65	7.25	5.71	0.31	7.75	5.70	0.11	7.5	5.69	−0.05	7.75	5.75	0.92	7.5	5.74	0.84	8	5.73	0.52
	1	0	5.64	0	5.60	−0.76	0	5.74	1.75	0	5.68	0.66	0	5.66	0.31	0	5.65	0.11	0	5.64	−0.05	0	5.70	0.93	0	5.69	0.85	0	5.67	0.53
	2	0	5.59	0	5.54	−0.77	0	5.68	1.76	0	5.62	0.67	0	5.60	0.31	0	5.59	0.11	0	5.58	−0.05	0	5.64	0.94	0	5.63	0.86	0	5.62	0.53
3	0	10	5.52	10	5.53	0.06	9.75	5.47	−0.87	10	5.44	−1.52	10	5.55	0.53	10	5.53	0.23	10	5.50	−0.42	10	5.55	0.55	10	5.54	0.40	10	5.55	0.48
	1	0	5.47	0	5.48	0.06	0	5.43	−0.87	0	5.39	−1.53	0	5.50	0.53	0	5.49	0.23	0	5.45	−0.43	0	5.50	0.55	0	5.50	0.41	0	5.50	0.48
	2	0	5.41	0	5.42	0.06	0	5.36	−0.88	0	5.33	−1.55	0	5.44	0.54	0	5.43	0.24	0	5.39	−0.43	0	5.44	0.56	0	5.43	0.41	0	5.44	0.49
	3	0	5.35	0	5.35	0.06	0	5.30	−0.90	0	5.27	−1.57	0	5.38	0.54	0	5.36	0.24	0	5.33	−0.44	0	5.38	0.57	0	5.37	0.42	0	5.38	0.49
4	0	10	5.64	10	5.70	0.94	10	5.70	0.94	10	5.64	−0.04	9.75	5.45	−3.43	10	5.56	−1.44	10	5.66	0.30	10	5.56	−1.45	10	5.67	0.49	10	5.64	−0.05
	1	0	5.54	0	5.59	0.95	0	5.59	0.95	0	5.54	−0.04	0	5.35	−3.49	0	5.46	−1.47	0	5.56	0.31	0	5.46	−1.48	0	5.57	0.50	0	5.54	−0.05
	2	0	5.42	0	5.47	0.98	0	5.47	0.98	0	5.42	−0.04	0	5.23	−3.57	0	5.34	−1.50	0	5.44	0.31	0	5.34	−1.51	0	5.45	0.52	0	5.42	−0.05
5	0	15	5.11	15	5.21	2.02	15	5.25	2.78	15	5.22	2.18	15	5.18	1.31	15	5.14	0.54	15	5.12	0.20	15	5.18	1.31	15	5.05	−1.19	15	5.16	0.87
	1	0	4.98	0	5.08	2.07	0	5.12	2.86	0	5.09	2.23	0.5	3.59	−27.77	0	5.00	0.55	0	4.99	0.20	0	5.04	1.35	0	4.92	−1.22	0	5.02	0.89
	2	0	4.80	0	4.90	2.15	0	4.94	2.96	0	4.91	2.32	0.25	3.11	−35.23	0	4.83	0.57	0	4.81	0.21	0	4.86	1.40	0	4.74	−1.26	0	4.84	0.92
6	0	15	5.22	15	5.31	1.65	15	5.31	1.60	15	5.28	1.14	15	5.27	0.79	15	5.28	1.00	15	5.32	1.76	15	5.31	1.65	15	5.33	2.03	15	5.25	0.42
	1	0	4.98	0	5.06	1.73	0	5.06	1.67	0	5.04	1.19	0	5.02	0.83	0	5.03	1.05	0	5.07	1.85	0	5.06	1.73	0	5.08	2.13	0.5	3.35	−32.68
	2	0	4.64	0	4.73	1.85	0	4.72	1.80	0	4.70	1.28	0	4.68	0.89	0	4.69	1.12	0	4.73	1.98	0	4.73	1.86	0	4.75	2.28	0.5	2.83	−38.97
	3	0	4.28	0.25	3.72	−13.12	0.5	3.31	−22.75	0.25	3.61	−15.61	0	4.32	0.97	0	4.33	1.22	0.25	3.79	−11.48	0	4.37	2.01	0	4.39	2.47	0.25	2.22	−48.26
7	0	20	4.85	20	4.80	−1.06	20	4.87	0.29	20	4.88	0.52	20	4.83	−0.39	20	4.82	−0.79	20	4.82	−0.63	20	4.73	−2.52	20	4.75	−2.10	20	4.91	1.10
	1	0	4.61	0	4.55	−1.11	0	4.62	0.31	0	4.63	0.55	0	4.59	−0.41	0	4.57	−0.83	0	4.57	−0.67	0	4.48	−2.66	0	4.50	−2.21	0	4.66	1.16
	2	0	4.34	0	4.29	−1.18	0	4.35	0.33	0	4.37	0.58	0	4.32	−0.43	0	4.30	−0.88	0	4.31	−0.71	0	4.22	−2.82	0	4.24	−2.34	0	4.39	1.23
8	0	20	4.96	20	4.97	0.21	20	5.07	2.11	20	5.04	1.63	20	4.98	0.44	20	4.96	−0.06	20	5.07	2.18	20	5.05	1.88	20	5.07	2.29	20	4.96	0.03
	1	0	4.74	0	4.75	0.22	0	4.84	2.21	0	4.82	1.71	0	4.76	0.46	0.75	2.58	−45.49	0.25	3.50	−26.07	0	4.83	1.97	1	2.14	−54.86	1	2.39	−49.52
	2	0	4.46	0	4.47	0.23	0	4.56	2.35	0	4.54	1.81	0.5	2.91	−34.77	1	2.09	−53.18	0.75	2.20	−50.63	0	4.55	2.09	1.5	1.45	−67.60	0.75	1.90	−57.39
	3	0	4.28	0	4.29	0.24	0	4.38	2.44	0	4.36	1.89	1	2.32	−45.73	1	1.63	−61.94	0.5	2.02	−52.76	0	4.37	2.18	0.5	1.48	−65.41	0.75	1.58	−63.04
Average		3.81	5.17	3.81	5.16	−0.12	3.82	5.18	0.21	3.80	5.17	−0.03	3.86	4.92	−5.42	3.91	4.90	−5.93	3.86	4.94	−5.02	3.81	5.19	0.44	3.91	4.87	−6.71	3.95	4.70	−10.36

Note: Gap=[(fit from other instability threshold)−(fit from instability threshold $\theta = 0.1$)]×100/(fit from instability threshold $\theta = 0.1$).

8.3. Sensitivity analysis on the instability threshold

To analyze the influence of instability threshold θ on the rescheduling problem, we vary the threshold between 0.2 and 1 with step size 0.2 (other three parameters are defined as: $\gamma_0 = 0.6$, $\gamma_c = 0.2$ and $\lambda = 10$). The calculated means and standard deviations of eight test instances are reported in Appendix N, and then the fitness value (i.e., fit) and corresponding gap in each scheduling stage can be obtained, as displayed in Table 10. The nonparametric Kruskal–Wallis test with 90% confidence level is carried out on AN and fit values, and the test results show that the instability threshold has no significant effect on the rescheduling problem since the test p-values are all greater than 0.99 (similar conclusion can also be obtained from Fig. 22).

9. Conclusions and future work

This paper proposes a hybrid multi-objective evolutionary algorithm (HMOEA) to deal with an adaptive flexible job-shop rescheduling problem with real-time order acceptance (ROA) and condition-based preventive maintenance (CBPM). More specifically, a CBPM policy is first designed to find the optimal maintenance planning for each machine. Second, a multi-objective optimization model is developed to characterize the concerned problem. Next, a HMOEA with hybrid initialization method, hybrid local search operators and adaptive rescheduling strategies is proposed to cope with the problem. Through numerical simulation, we discover that (1) the proposed CBPM policy with non-periodic concave decreasing inspection interval is better; (2)

the order sorting Rule 7 is the best one if a small-scale problem is solved; otherwise, the best one is Rule 14; (3) the improved operators and the proposed HMOEA algorithm are better than its peers; (4) different rescheduling strategies have significant differences in different evaluation indicators, and the optimal rescheduling strategies in different rescheduling scenarios are recommended; (5) ROA strategy is better than AOA strategy under limited production capacity, and the rejection penalty has a significant impact on these two order acceptance strategies; and (6) two initial addition ratios and a instability threshold have no significant effect on the adaptive rescheduling problem.

As for the future research on this topic, we aim at (1) studying the customer's order acceptance and scheduling (COAS) problem, i.e., each customer may have multiple orders; (2) considering new (re)scheduling environments and factors for ROA, e.g., sequence-dependent setup times and energy consumption in distributed flexible workshop/hybrid flowshop (Meng et al., 2022, 2020, 2019); (3) analyzing the impact of learning effect, forgetting effect, job deterioration effect and maintenance or due date time window on the integrated optimization; and (4) exploring the coupling relationship among production scheduling, CBPM and maintenance resource assignment in a dynamic environment.

CRediT authorship contribution statement

Youjun An: Investigation, Methodology, Writing – original draft. **Xiaohui Chen:** Supervision, Funding acquisition. **Kaizhou Gao:** Supervision, Funding acquisition, Writing – review & editing. **Lin Zhang:** Methodology, Writing – review & editing. **Yinghe Li:** Project administration, Writing – review & editing. **Ziye Zhao:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.eswa.2022.118711.

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